ALGORITHMIC BIAS AND FEEDBACK:
BIAS ON THE WEB

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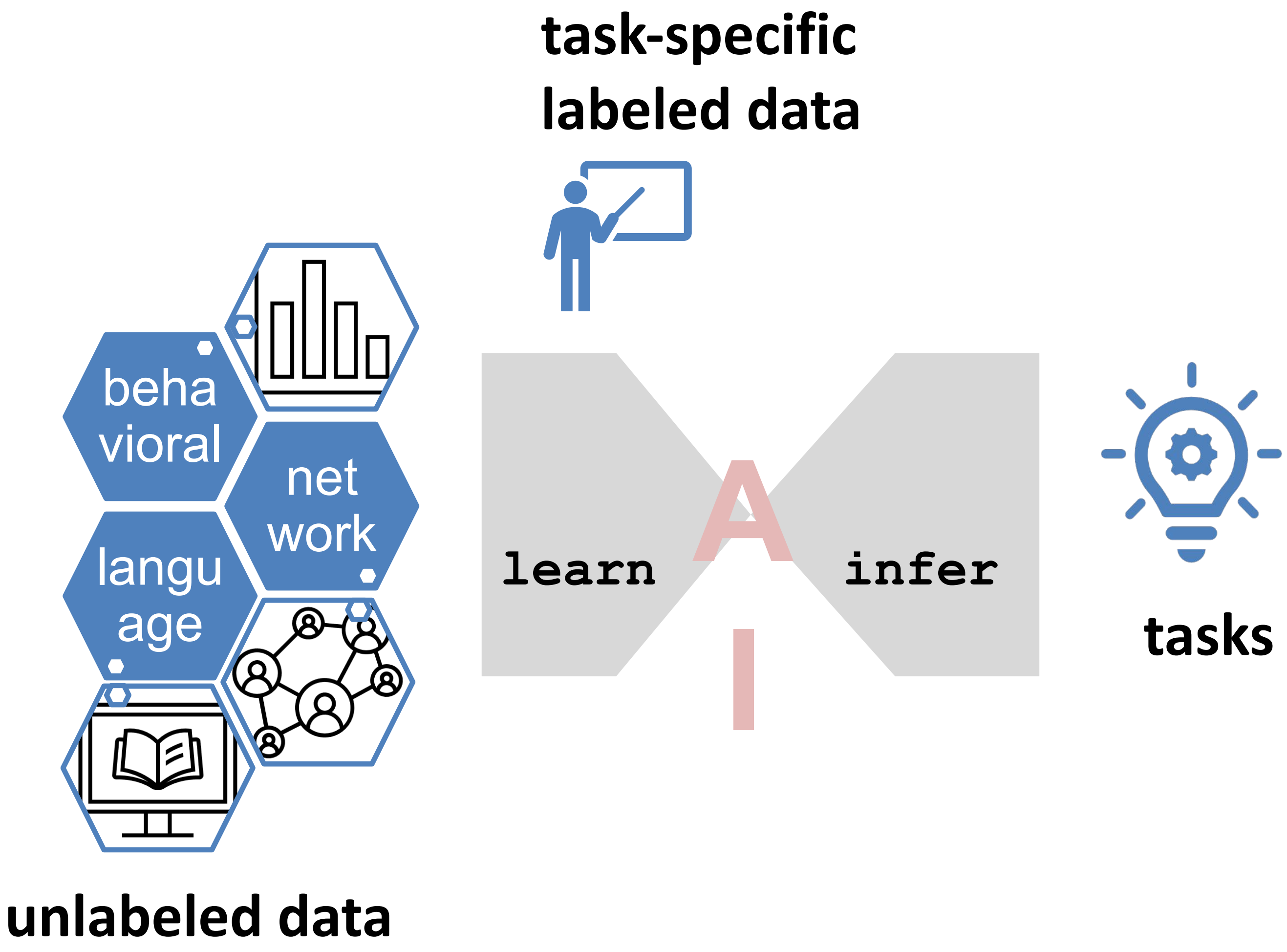
DSCI 531

AI pipeline



task-specific
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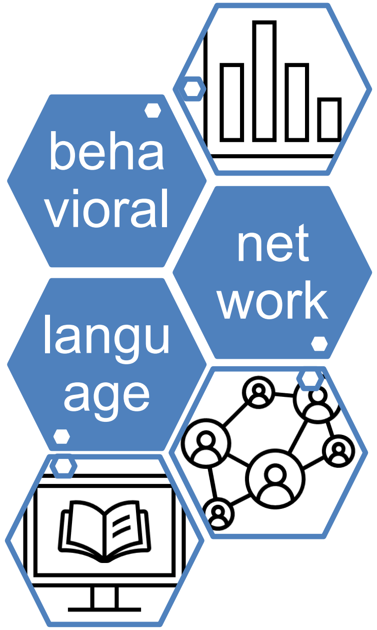
tasks

AI pipeline: feedback loops

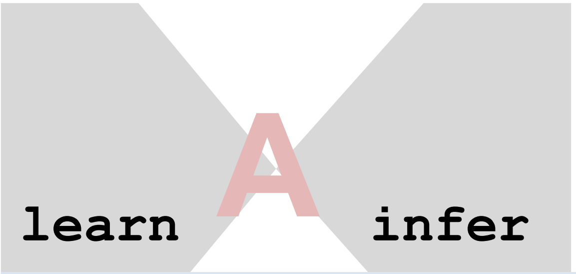
Data is biased
- Cognitive biases
- Psychological biases: negativity, ingroup favoritism, …
- Social biases: cultural biases & Stereotypes: body image, emotionality, moral values, …
- Demographic: gender, ideology, …

Data collection is biased
- Incomplete
- Non-representative
- …



biased
~~unlabeled~~ **data**

task-specific labeled data **is noisy, biased, subjective**



tasks

algorithms are biased
- representation bias
- defining outcomes & features …

feedback loops amplify bias

Risks of bias
- AI models learned from biased data can unfairly discriminate
- People interact with AI outputs, they create more biased data
- This creates feedback loops that risk amplifying biases

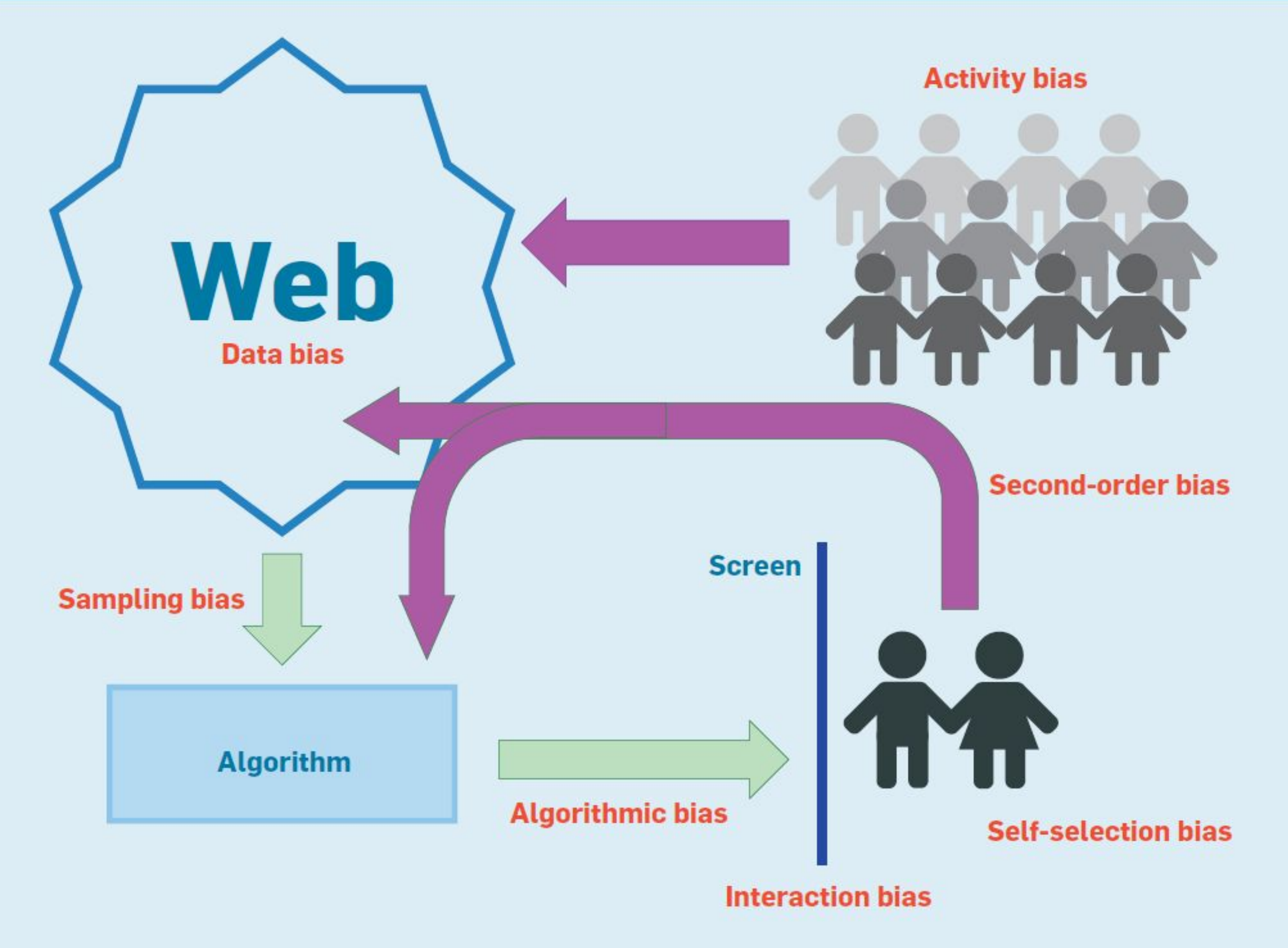
Bias on the Web

[Baeza-Yates, R. (2018). Bias on the web. *Communications of the ACM*, *61*(6), 54-61.]

- **Any remedy for bias starts with awareness of its existence.**
- **Bias on the Web reflects biases within ourselves, manifested in subtler ways.**
- **We must consider and account for bias in the design of Web-based systems that truly address the needs of users.**

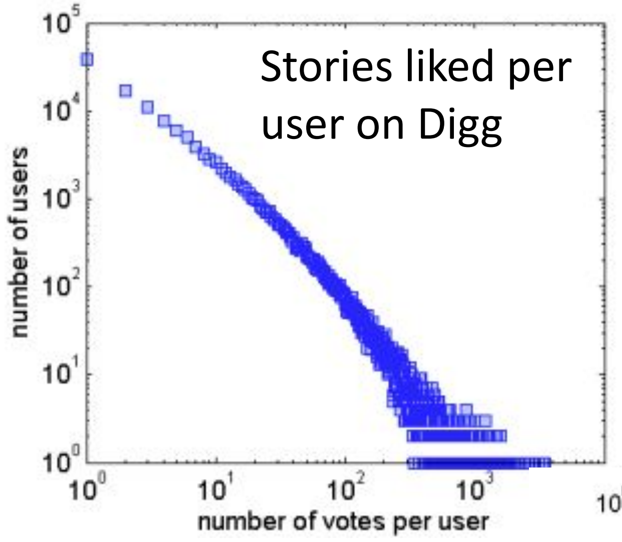
[Baeza-Yates, R. (2018). Bias on the web. *Communications of the ACM*, *61*(6), 54-61.]

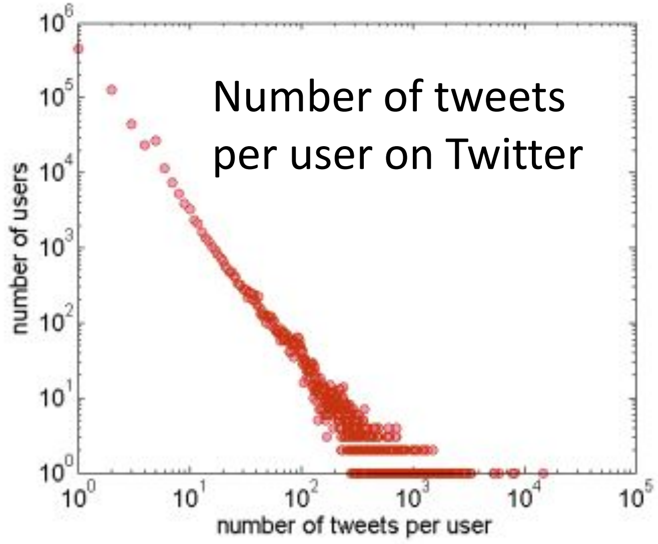
Figure 1. The vicious cycle of bias on the Web.



- Every decision and interaction creates an opportunity for bias to creep in
- Biases compound

Online engagement is highly heterogeneous





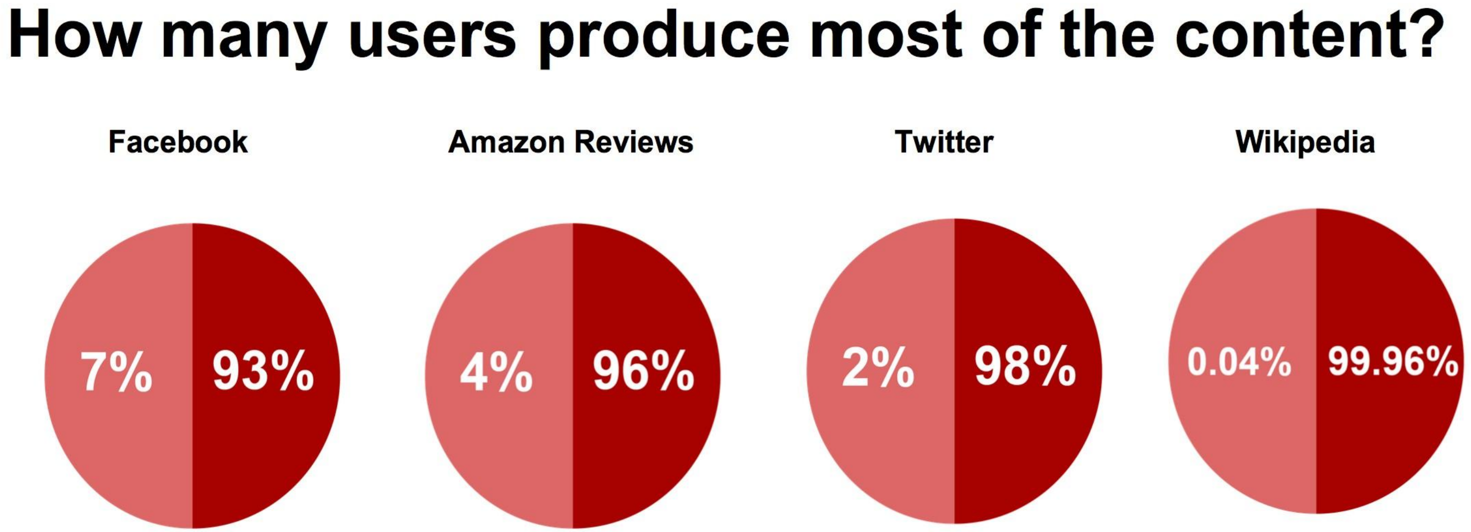
- Only a few users are very active – most are inactive
- Typically, activity (number of likes/tweets/retweets per user) has an L-shaped form
  - Use log-log plot to visualize (plots on the left)
  - Indicative of high inequality of activity

Activity Bias, or Wisdom of a Few

Link bias

- half of the Twitter users in the dataset were following only a few select celebrities (0.05% of the most popular people attracted almost 50% of all participants);
- Inequality of popularity (similar to wealth inequality)

Activity bias



“digital desert”

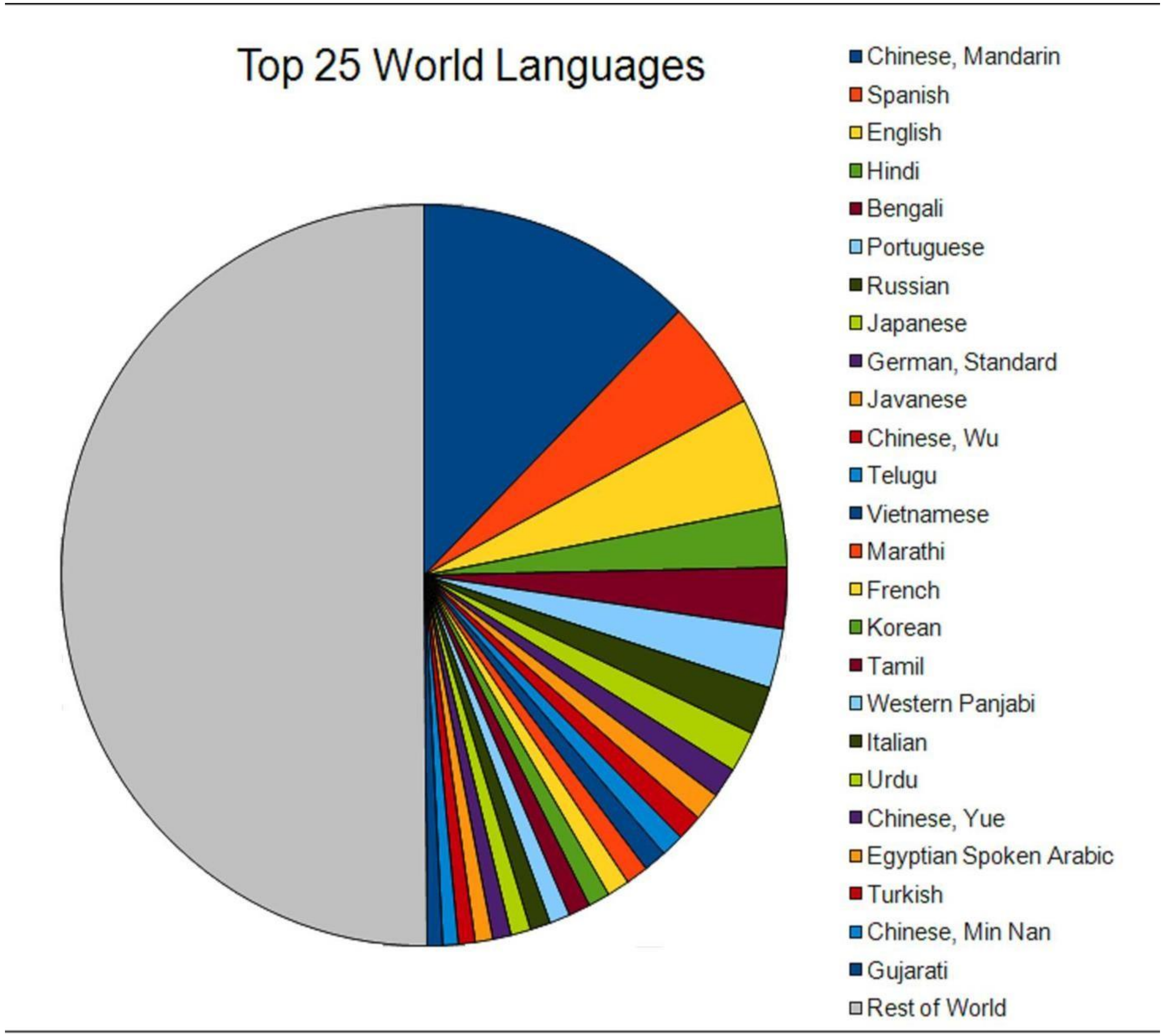
- Activity bias generates a “digital desert” across the Web, or Web content no one ever sees.
- 1.1% of the tweets were written by people without followers.
- 31% of the Wikipedia articles added or modified in May 2014 were never visited in June.
- The size of the digital desert on the Web is in the 1% to 31% range.

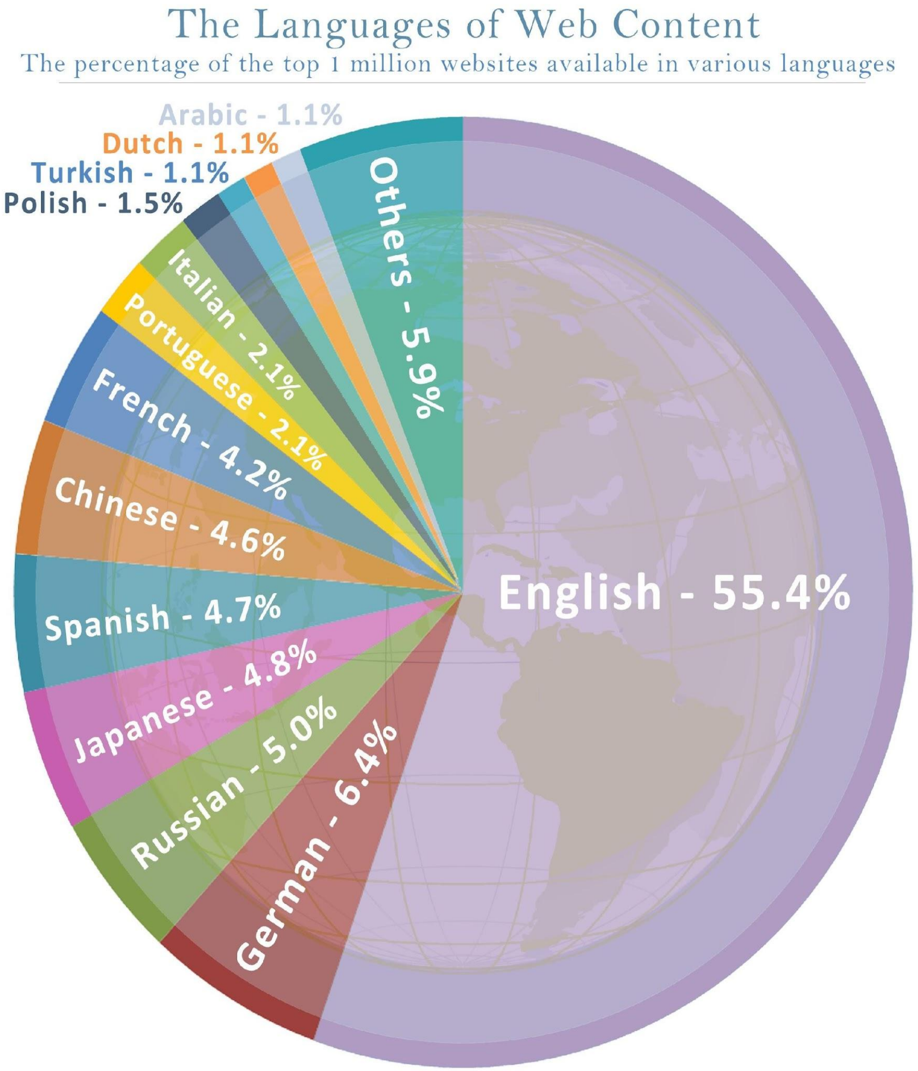
Is bias bad?

The Web works thanks to bias

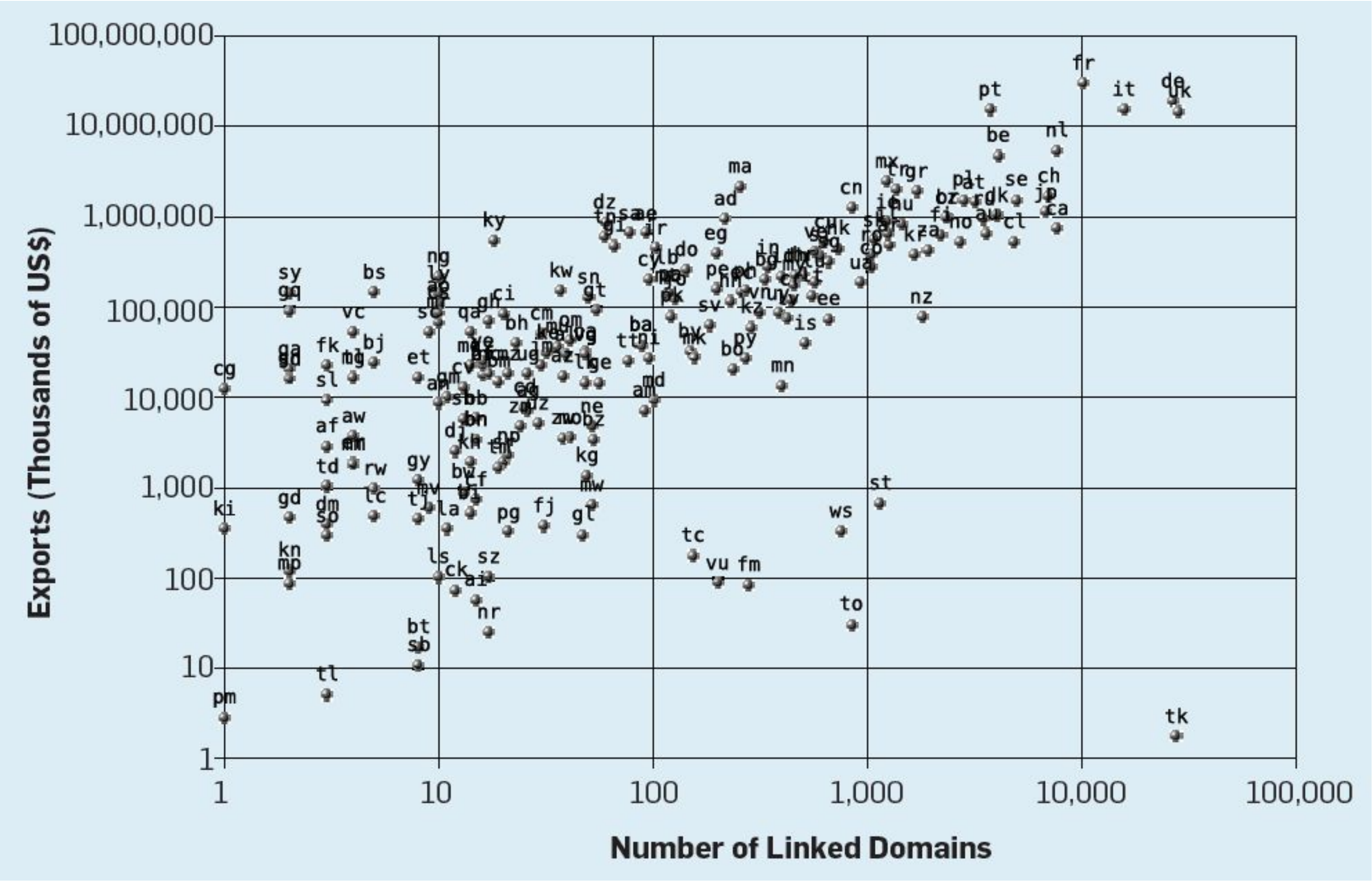
- Web traffic
  - Local caching
  - Proxy/Akamai caching
- Search engines
  - Answer caching
  - Essential web pages
    - 25% queries can be answered with less than 1% of the URLs [Baeze-Yates, Boldi, Chierichetti, WWW 2015]
- E-commerce
  - Most revenue comes from few items

Linguistic bias

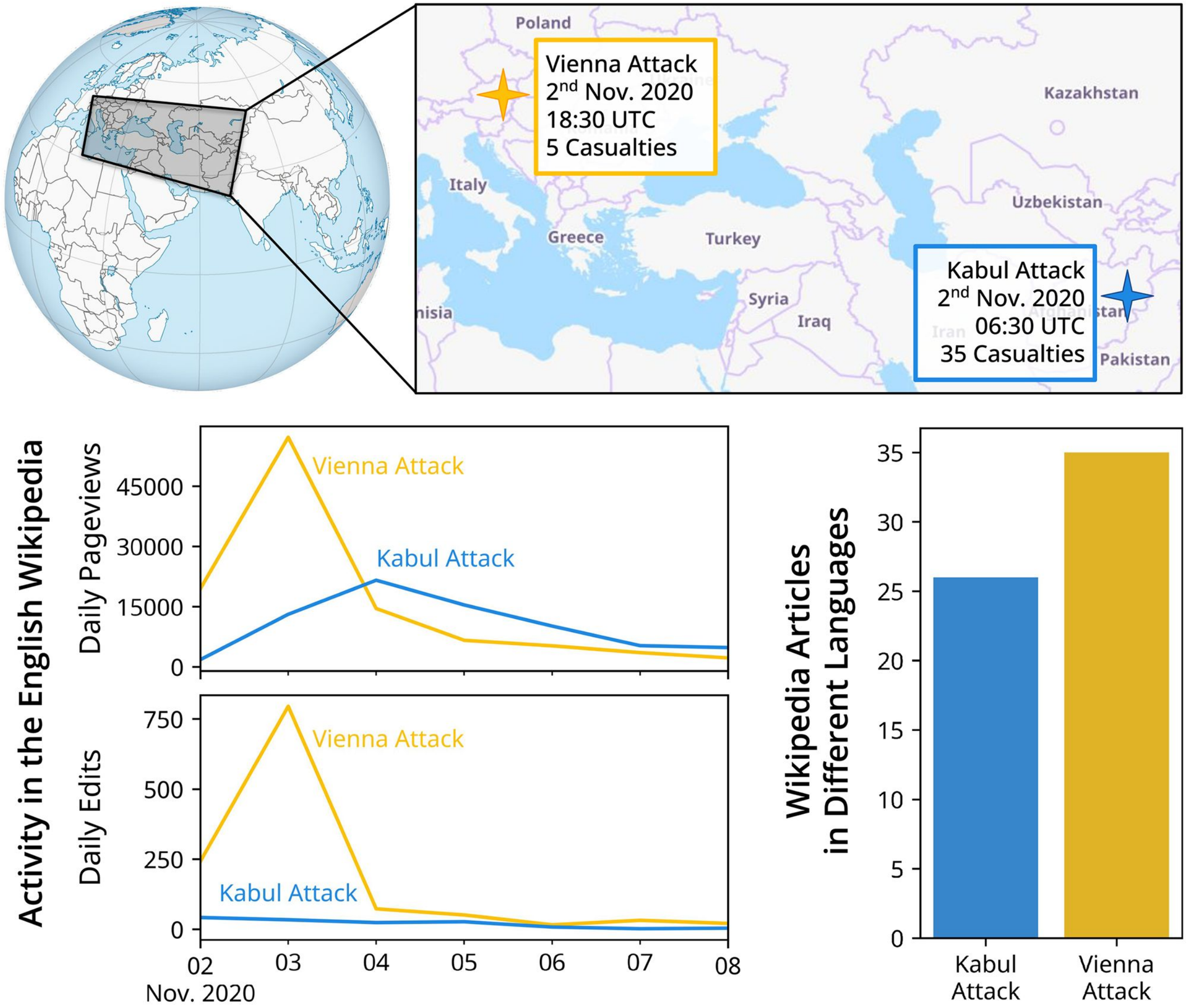




Geographic bias

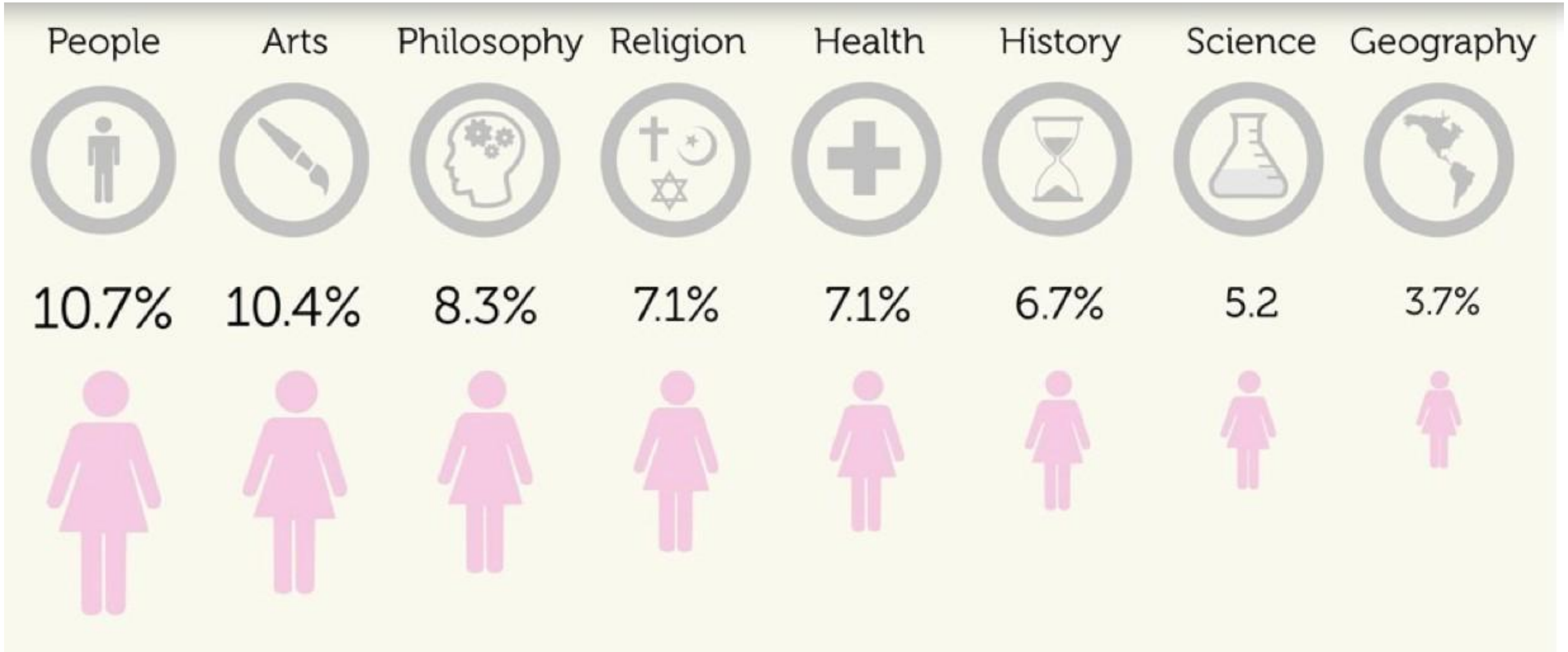


Wealth and regional disparities in attention to events on Wikipedia



Ruprechter T, Burghardt K, Helic D (2023) Poor attention: The wealth and regional gaps in event attention and coverage on Wikipedia. PLoS ONE 18(11): e0289325. https://doi.org/10.1371/journal.pone.0289325

Gender bias in Wikipedia



bias in Wikipedia biographies

[E. Graells-Garrido et al,. "First Women, Second Sex: Gender Bias in Wikipedia", ACM Hypertext'15]

Interaction between biases in content creation

- Systematic gender bias through history
- But, content creation bias too
  - less than 12% of Wikipedia biography editors are women
  - Worse gender bias in other categories (e.g., 4% of geography editors are women)
- 70% of influential journalists in the U.S. are men,
- Algorithms learning from news articles are thus learning from texts with systemic gender bias.
  - There could be additional cultural and cognitive biases

- Feedback loops amplify bias!

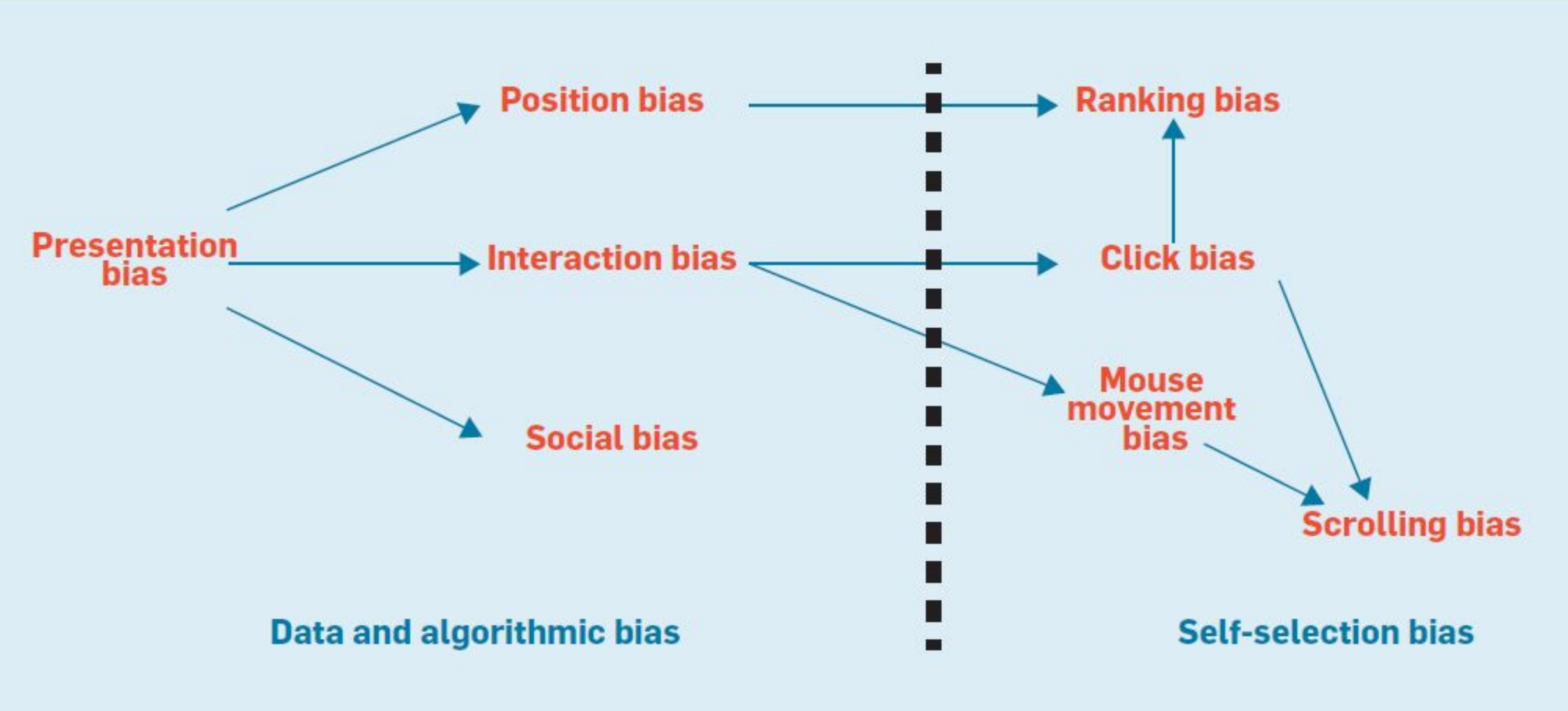
Platforms & biases

- User interactions on websites will be biased to what is shown
- In search systems top ranked results will get more clicks than other results
  - Ranking bias
    - The top-ranked result will thus attract more clicks than the others because it is both the most relevant and also ranked in the first position.
    - To avoid ranking bias, Web search engines need to de-bias click distribution
    - Otherwise, the popular pages become even more popular.
  - Interaction bias
- In recommender systems, items recommended will get more clicks than items not recommended
  - Vast amount of content not shown to users
- …

Biases in the user interface

Cascading biases

Figure 6. Dependency graph of biases affecting user interaction.



Cascading bias: Bias begets bias

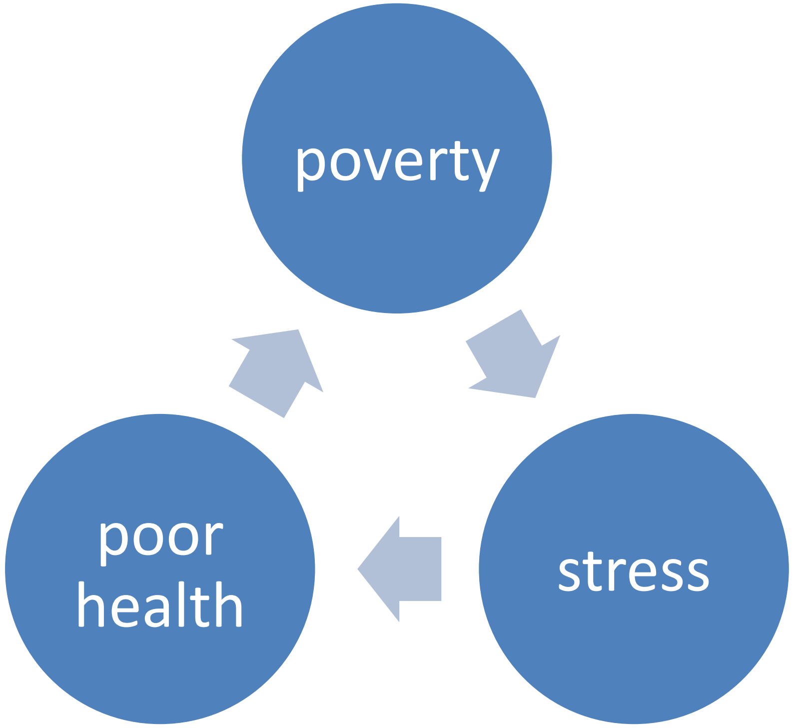
- Personalization algorithms (aka the filter-bubble effect),
  - If a personalization algorithm uses only our interaction data, we see only what we want to see, thus biasing the content to our own selection biases,
- Confirmation bias
  - We pay engage with content that is congruent with our pre-existing beliefs, feeding more biased information to personalization algorithms

- Creating <u>feedback</u> <u>loops</u> that amplify bias, a common mechanism across domains

A general mechanism for bias amplification

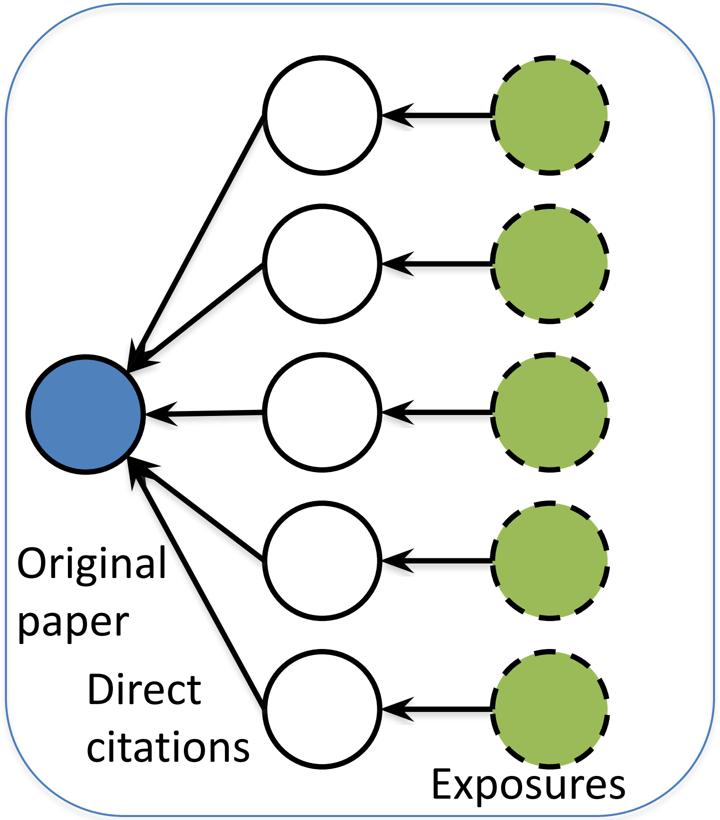
- **Positive feedback loops**: Advantage begets more advantage
- Cumulative advantage in science [Merton 1968]: Patterns of recognition are skewed in favor of the privileged
  - aka "the rich get richer and the poor get poorer"
  - "eminent scientists get disproportionately great credit for their contributions to science while relatively unknown scientists tend to get disproportionately little credit for comparable contributions."
- Also see preferential attachment in networks: a few nodes get most of the links [Barabasi & Albert 1999]

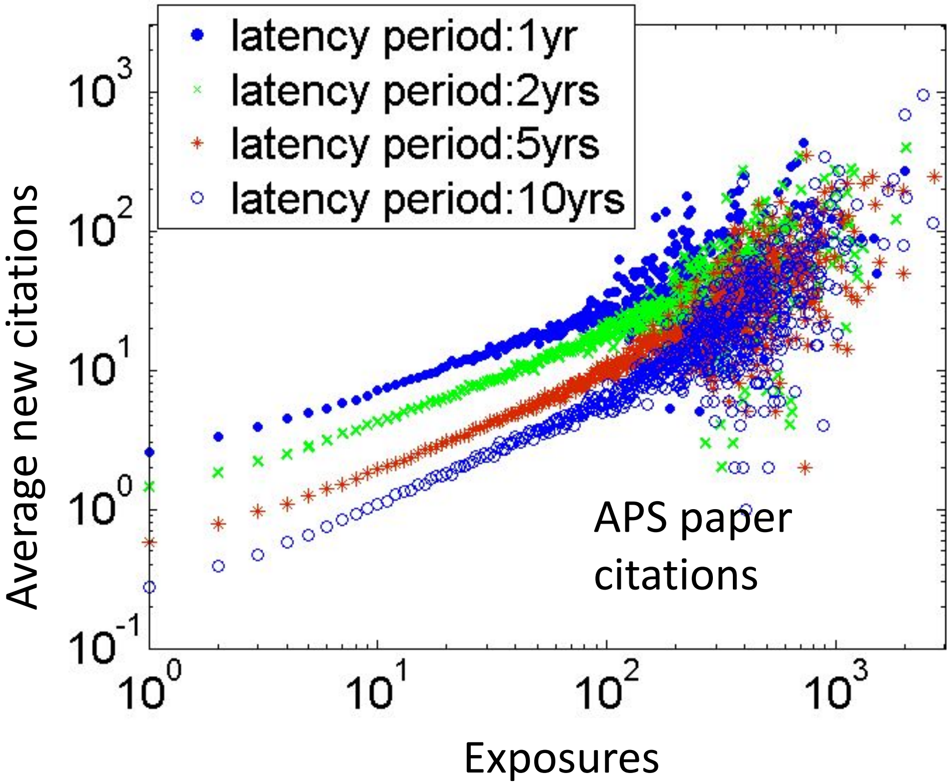
Negative feedback loops: Disadvantage begets more disadvantage



Cumulative advantage in science: citations

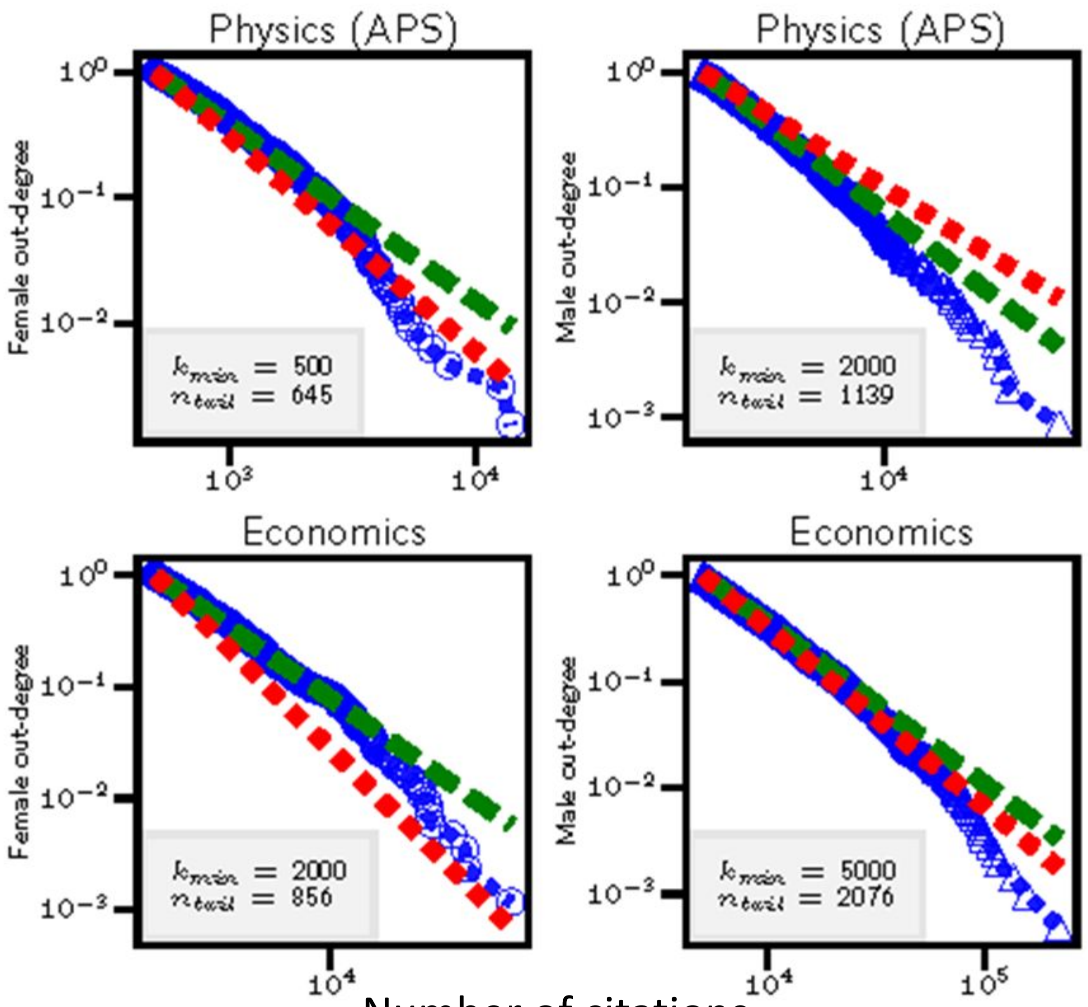
Citations beget more citations: each citation of a paper exposes to a wider audience, creating opportunities for others to find and cite it in their work.





Source: https://arxiv.org/abs/1710.00269

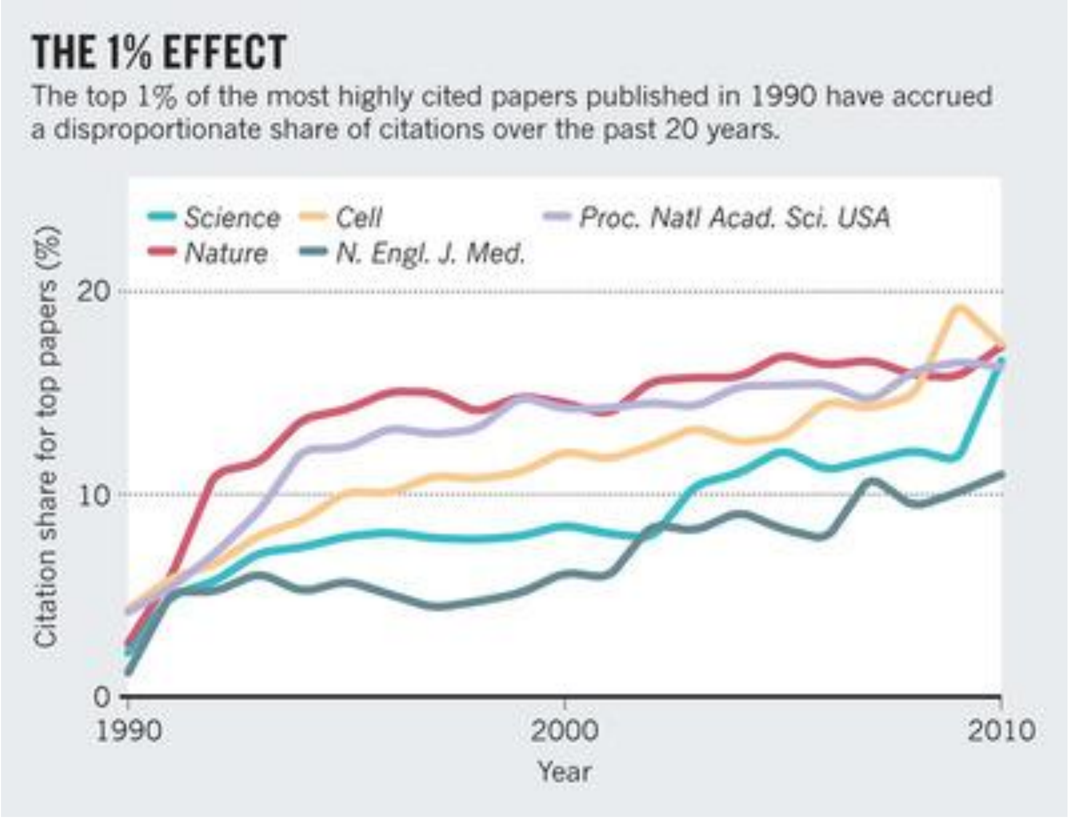
Inequality of citations



- The number of citations is <u>highly skewed</u>
  - Most researchers have only a few citations
  - A small number of researchers have large number of citations
  - Gini ~ 0.6-0.7 (*cf* US wealth inequality gini ~ 0.4)
- Similar distribution for the number of followers on Twitter, the number of "likes" tweets receive, etc.

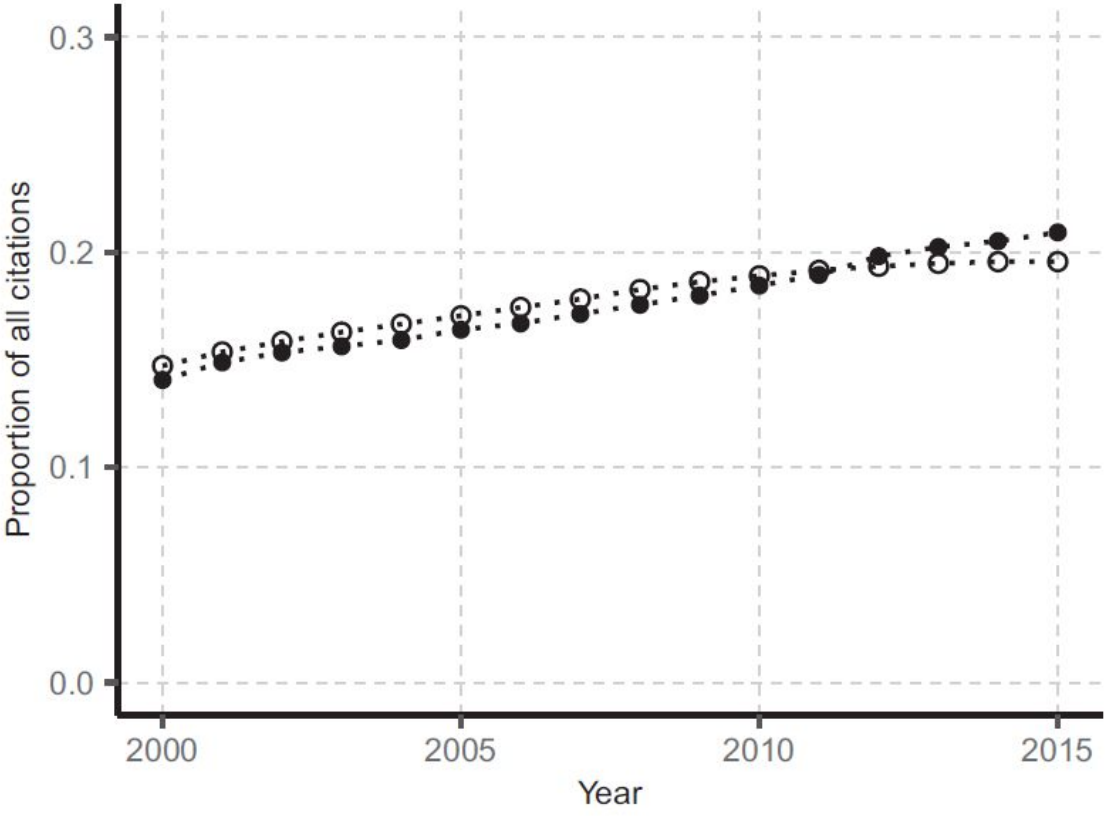
Inequality in science

A handful of papers dominates citations



Barabási, A. L., Song, C., & Wang, D. (2012). Handful of papers dominates citation. *Nature*, *491*(7422), 40-40.

A handful of authors dominate citations: Share of citations going to the top-1% most-cited authors



Nielsen, M. W., & Andersen, J. P. (2021). Global citation inequality is on the rise. *Proceedings of the National Academy of Sciences*, *118*(7).

COGNITIVE BIASES

Bounded rationality (aka "thinking is hard")

Herbert A. Simon

- **Bounded rationality**
- Constraints of available time, information, and cognitive capacity limit human ability to make rational decisions

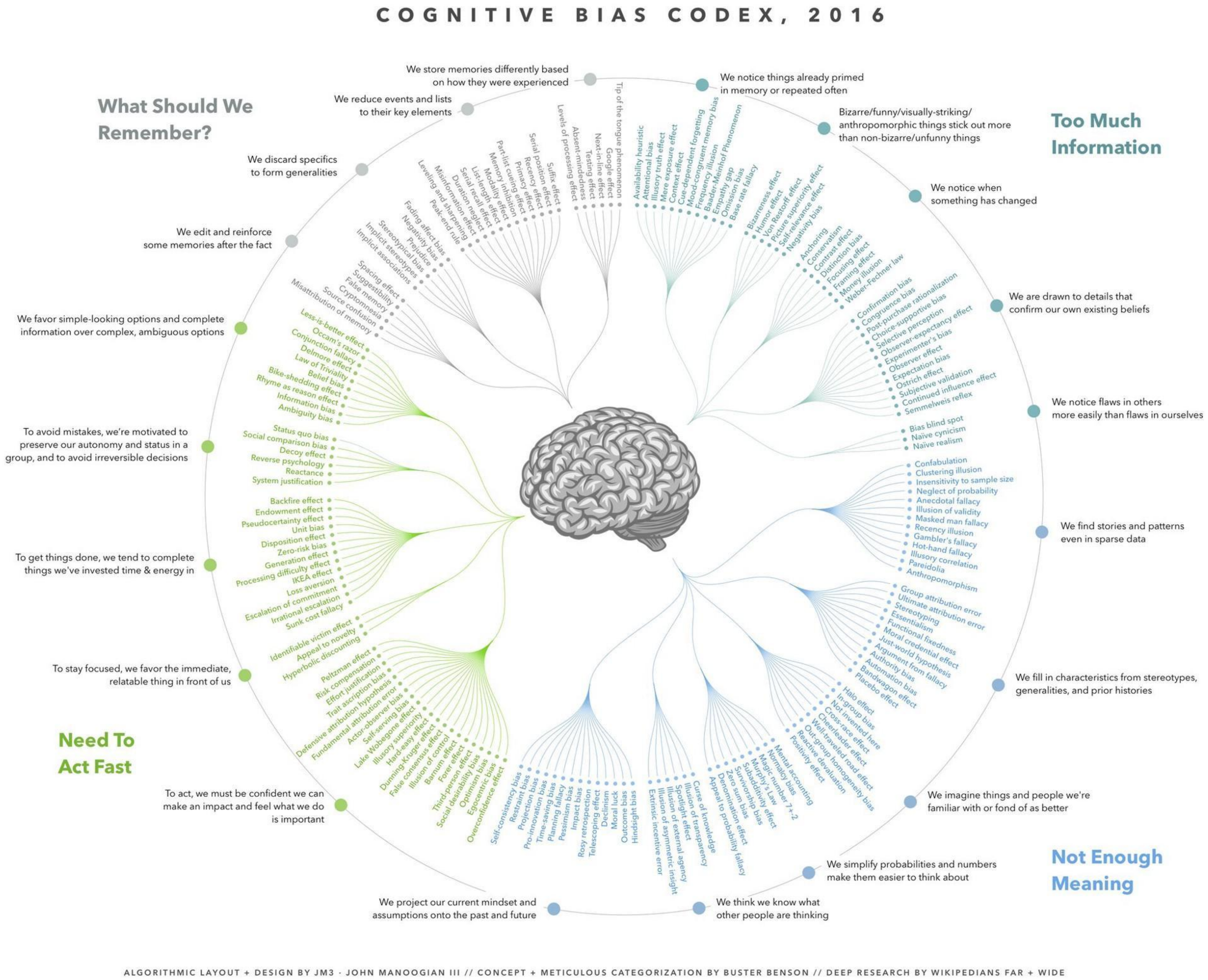
[Simon (1957). "A Behavioral Model of Rational Choice", in Mathematical Essays on Rational Human Behavior in a Social Setting.]

Daniel Kahneman

Amos Tversky

- **Heuristics and biases**
- Mental shortcuts that help people make quick, but less accurate decisions, by focusing brain's limited resources on the most salient information

[Tversy and Kahneman (1974). Judgment under uncertainty: Heuristics and biases. *Science*
Kahneman (2011) *Thinking Fast and Slow*.]



Informati

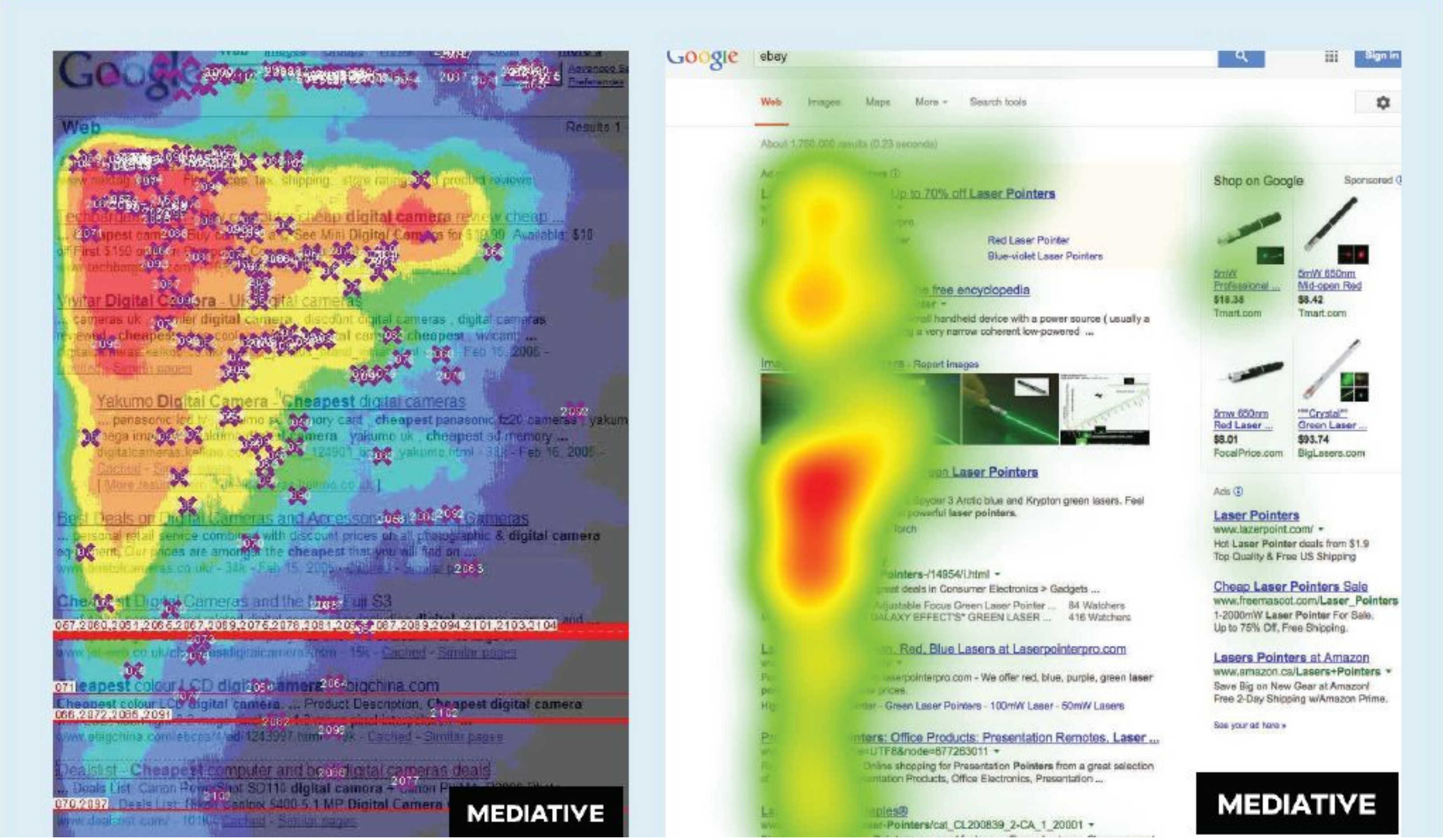
USC Viterbi School of Engineering

How do individual biases affect online behavior

Content near the top or near images attracts attention

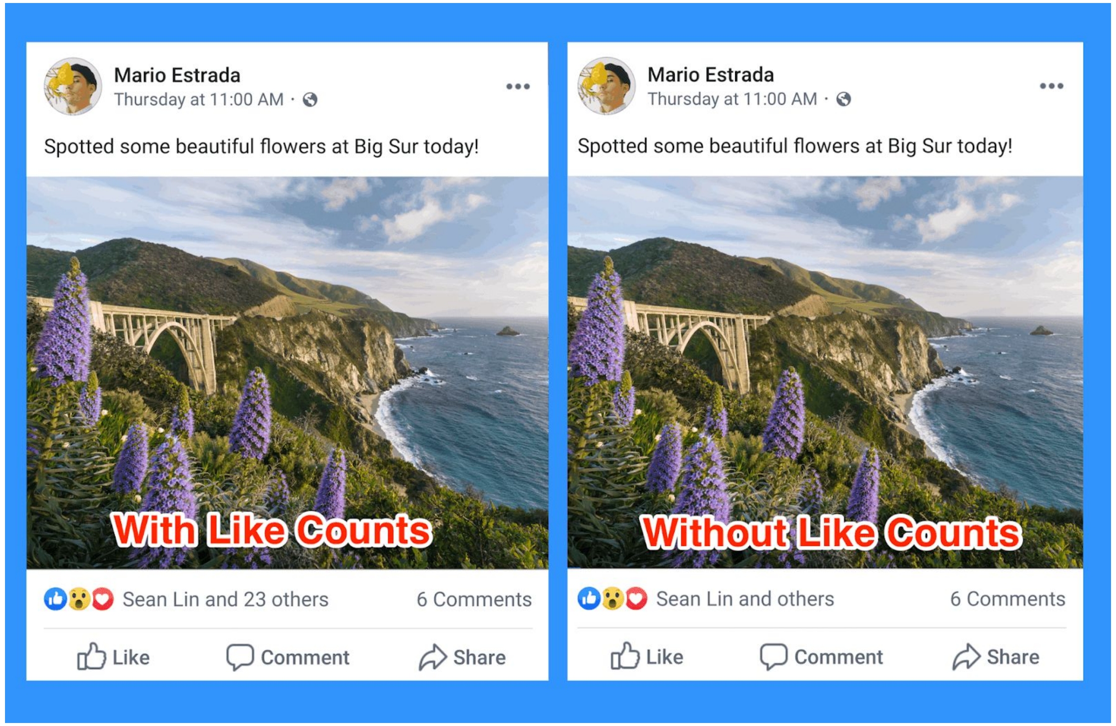
Position bias: People pay more attention to items at the top of the screen or a list of items [Payne 1951]

Figure 5. Heat maps of eye-tracking analysis on web-search results pages, from 2005 (left) to 2014 (right).[18]



Social proof bias

- Signals from other people affect our judgment.
  - "social conformity," "the herding effect", or "the bandwagon effect"



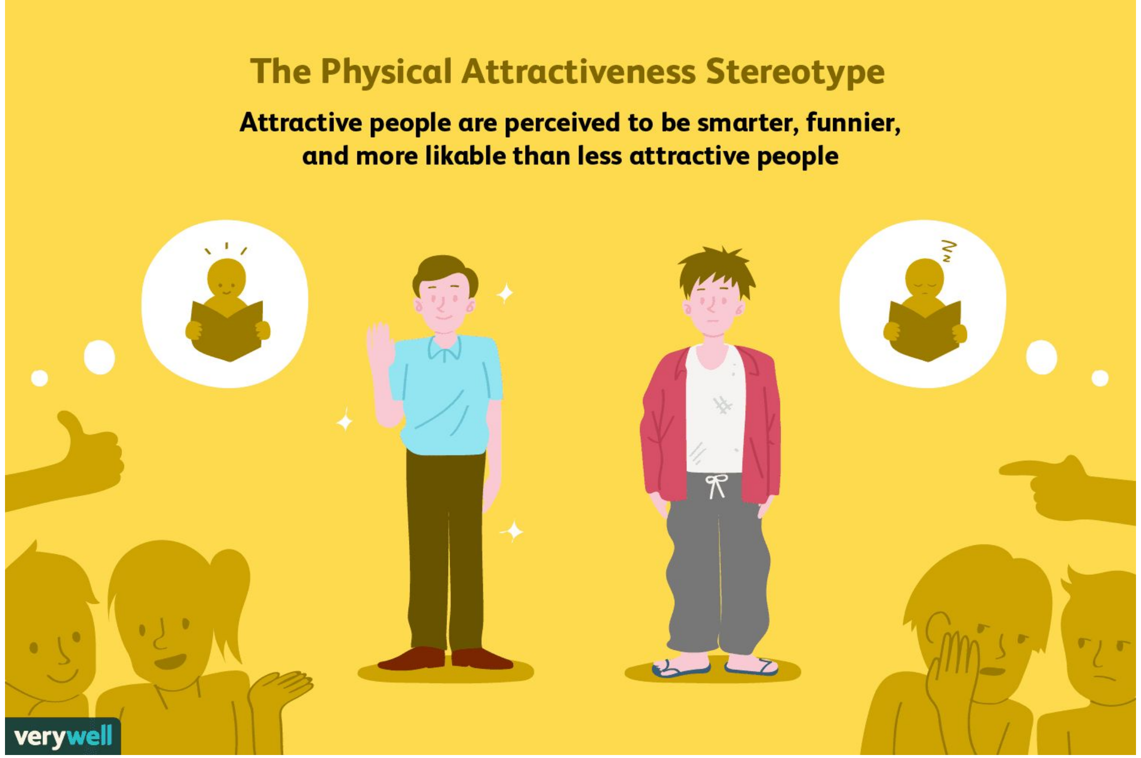
Negativity bias: Bad is stronger than good*

- Asymmetry in human perception
  - Criticism affects us more than praise; Negative first impressions are harder to overcome
  - Negative emotions have stronger impact than positive emotions
  - News coverage is mostly negative
- Negatively-biased credulity
  - Negative information (e.g., about threats) is more likely to be believed
  - Associated with ideology; conservatives more likely to believe negative information

* Baumeister, R. F., Bratslavsky, E., Finkenauer, C., & Vohs, K. D. (2001). Bad is stronger than good. *Review of general psychology*, 5(4), 323-370.

Halo effect

- Tendency for positive impressions of a person in one area to positively or negatively influence one's opinion or feelings in other areas.
- One trait of a person or a thing is used to make an overall judgment about them

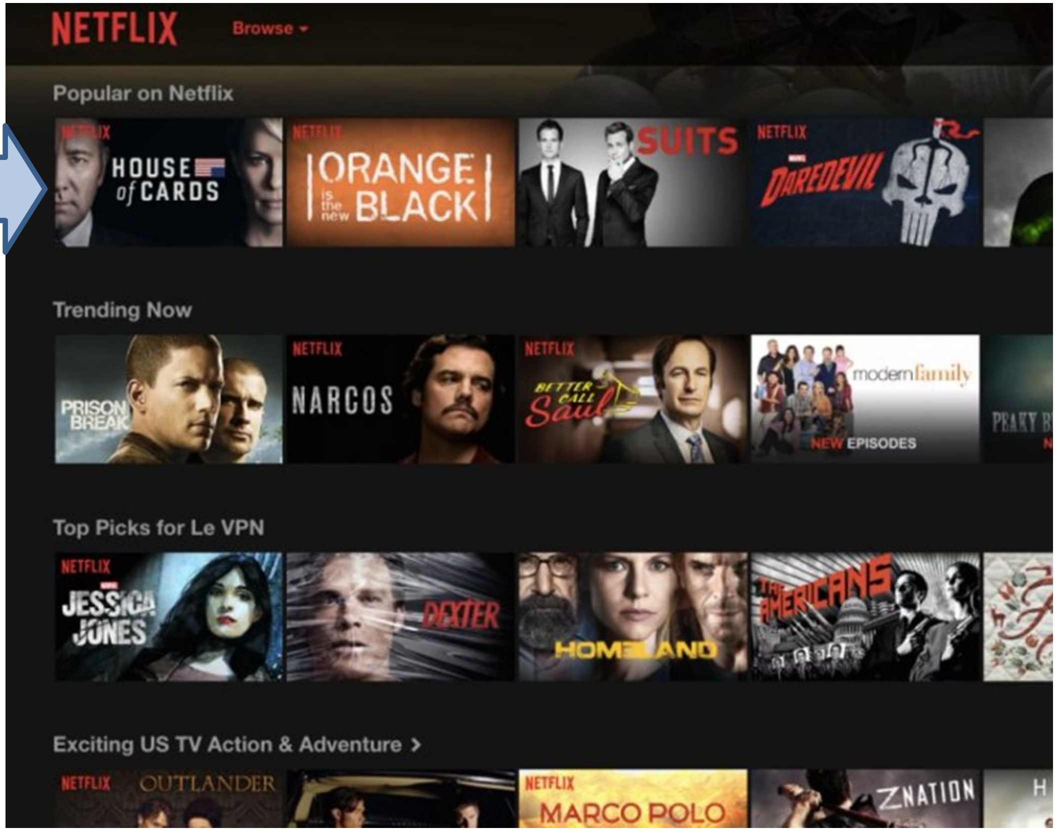


INTERACTIONS BETWEEN COGNITIVE BIASES & ALGORITHMS

Crowdsourced ranking

- **Crowdsourced ranking**
  - aggregates decisions of many people to rank items
  - "quality items float to the top"
- Instability in popularity ranking
  - Music Lab study [Salganik+ 2006]
  - Few items become hits; most languish in obscurity
  - Can't predict what items become hits
    - Harry Potter book was rejected 9 times

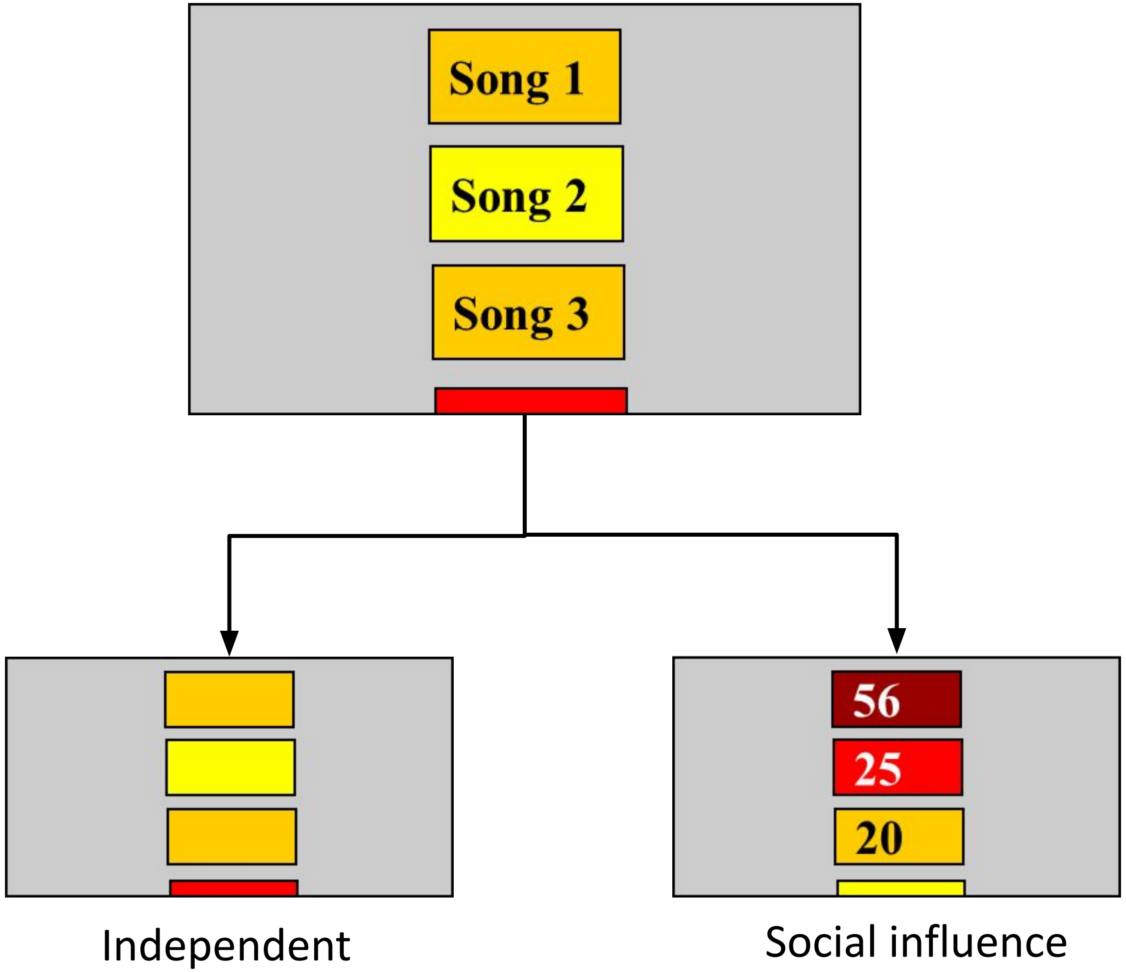
Popularity ranking: Popular on Netflix (or Twitter, Reddit, StackOverflow, …)



MusicLab study

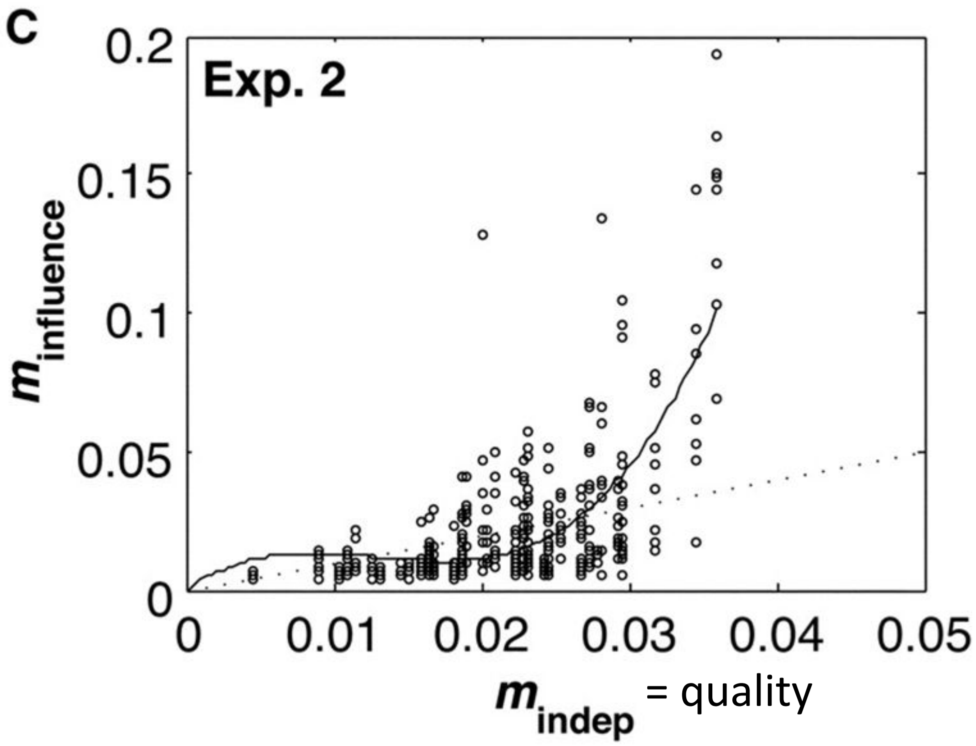
Matthew J. Salganik, Peter Sheridan Dodds, Duncan J. Watts. Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market. *Science* 10 Feb 2006 : 854-856

- Online experiment to simulate a cultural market
- Users shown a list of songs in one of two conditions
  - **Independent**: songs in random order
  - **Social influence**: songs ranked by popularity
- Users download songs they like
  - Popularity = fraction of all downloads a song receives
  - Quality = popularity in the independent condition



Popularity vs quality

Song market share*: social influence vs independent conditions

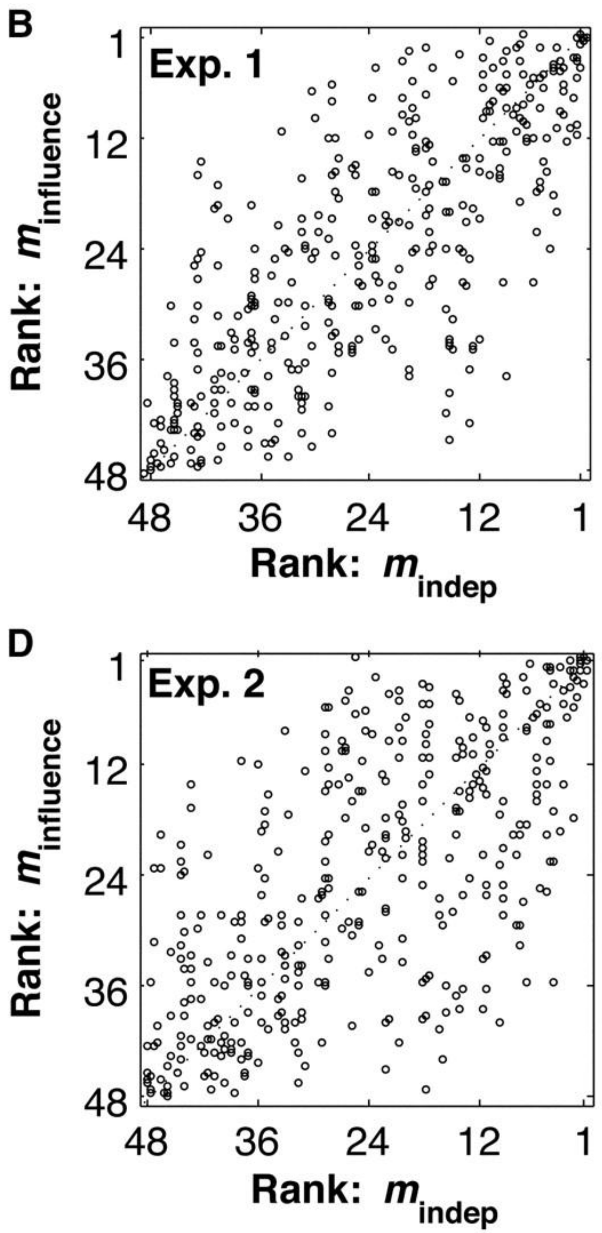


= quality

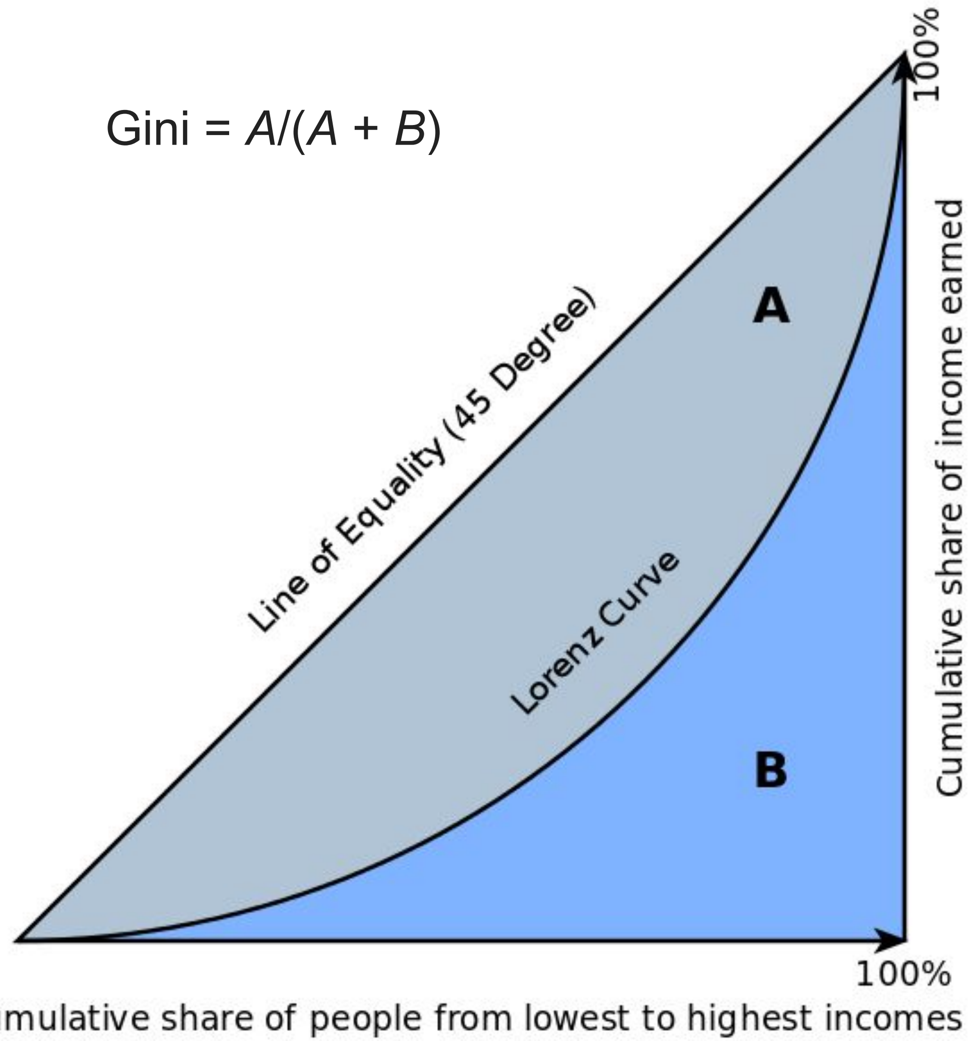
* market share = fraction of all downloads in an experiment attributed to a given song

Ranking instability

- **Huge differences in popularity (inequality)**
  – Few items become very popular
- **Popularity is weakly correlated with quality**
  – The best and worst songs were correctly ranked.
  – Rank of mediocre songs is unpredictable
- **Multiple worlds experiment**
  – Rerun the experiment from the same initial conditions. Different songs become popular



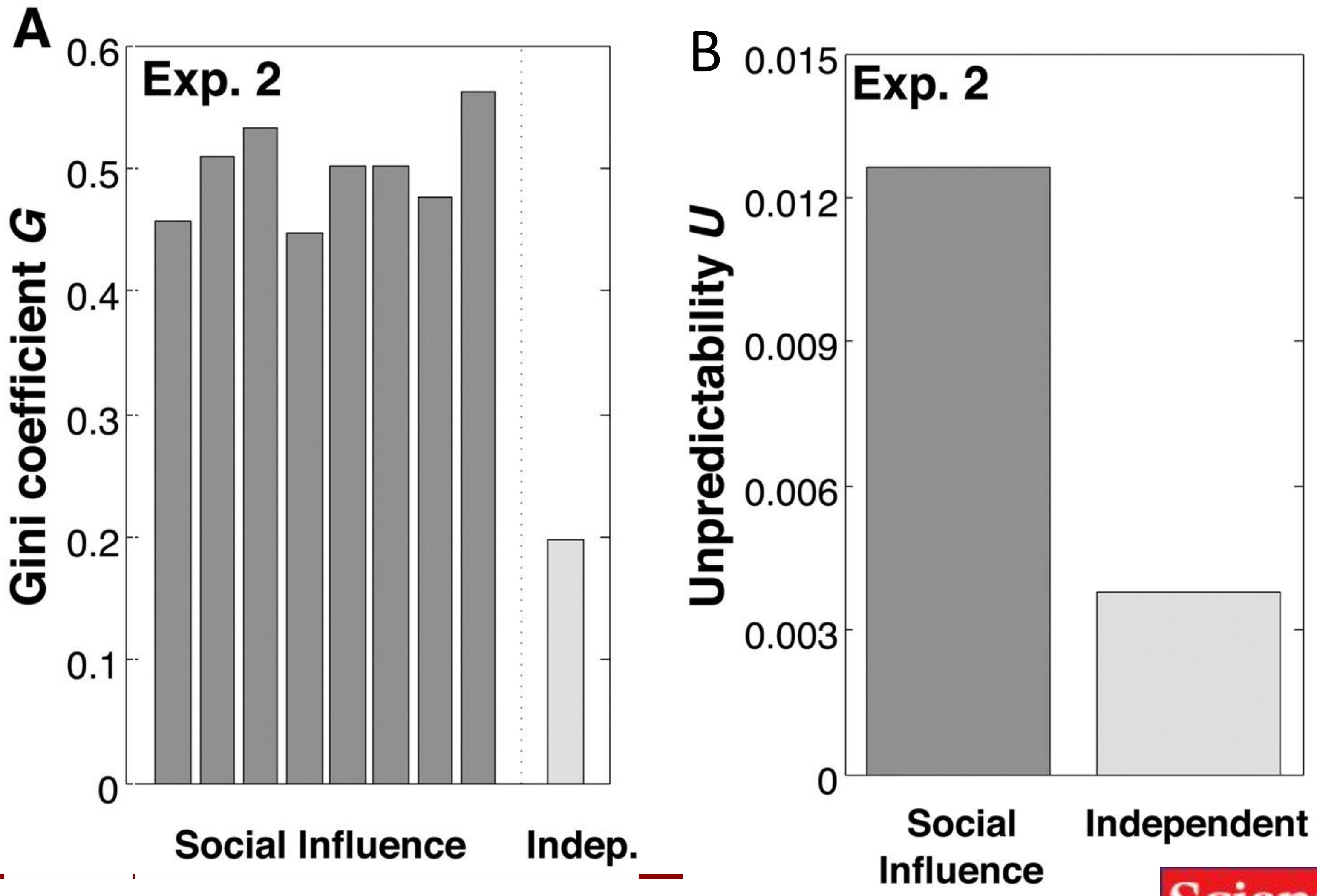
Measuring inequality



Gini coefficient measures the degree of inequality in a distribution of values.

$$G = \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i - x_j|}{2 \sum_{i=1}^{n} \sum_{j=1}^{n} x_j}$$

Inequality (A) and unpredictability (B) of song market share in the social influence vs independent conditions worlds



Matthew J. Salganik et al. Science 2006;311:854-856

Published by AAAS

Conclusions of the MusicLab study

- Social influence distorts collective outcomes in markets
  - Popularity is only weakly correlated with quality
  - Social influence is responsible for the inequality and unpredictability of popularity

But, MusicLab study did not attempt to control for the salience of the social signal – attention paid to songs solely due to their position in the list

Social influence or position bias?

- **Social influence**: People pay more attention to the popular choices

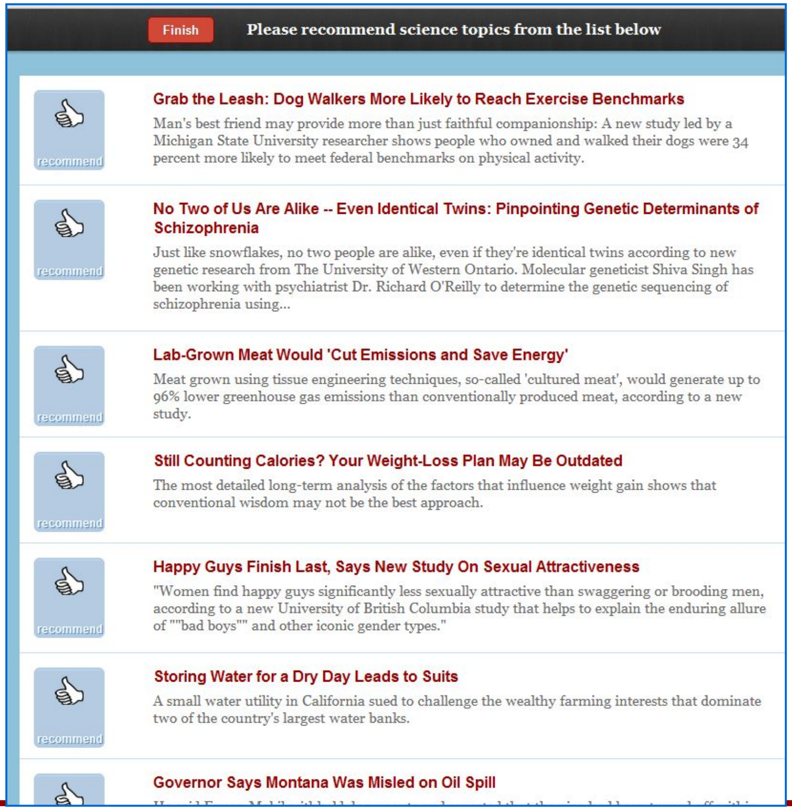
- **Position bias**: People pay more attention to items at the top of the screen or a list of items [Payne 1951]

[Buscher et al, CHI'09]

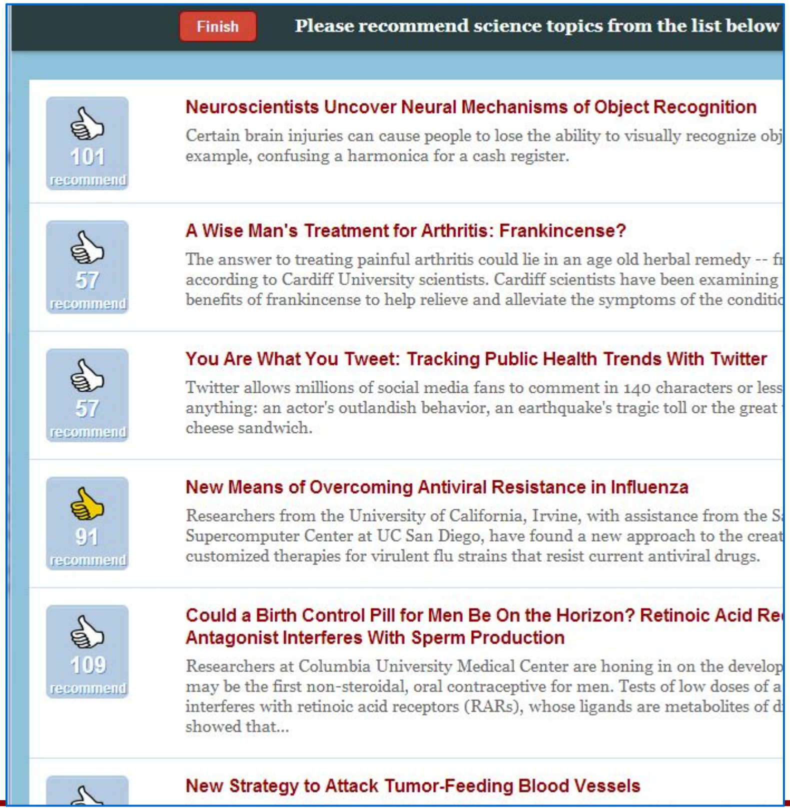
Measuring position bias

- Quantify the effects of position bias on individual's choices and collective outcomes (story popularity) via Amazon Mechanical Turk experiments

No social influence condition



Social influence condition (#likes shown)



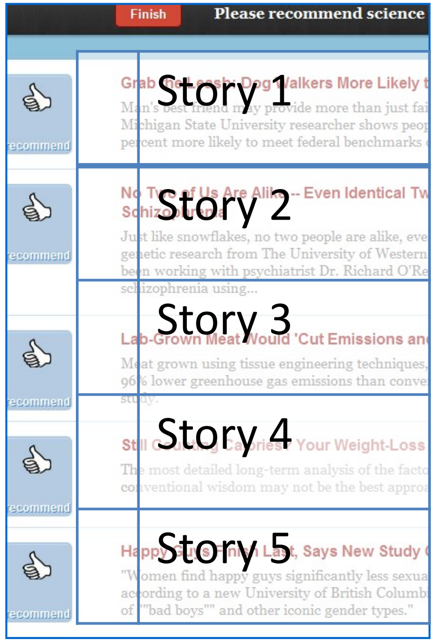
[Lerman & Hogg (2014), "Leveraging position bias to improve peer recommendation" in *PLoSOne*]

4 algorithmic ranking schemes

- Participants asked to recommend (like) stories from a list 100 science stories
  - click on URL to read the story
  - vote for (like) the story
- Vary order → measure outcomes (popularity)

[random order]

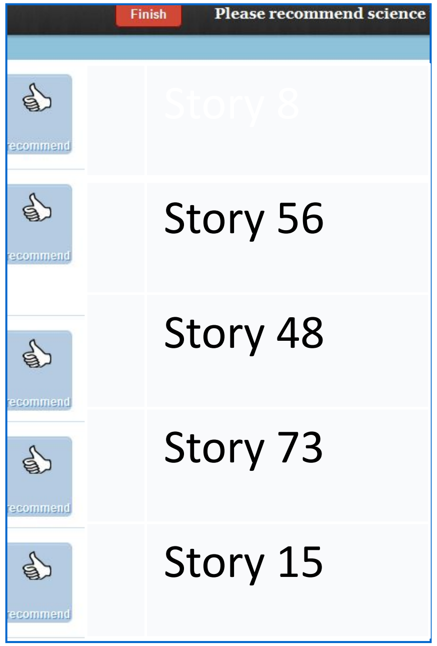
control



[fixed order]

[by popularity]

likes



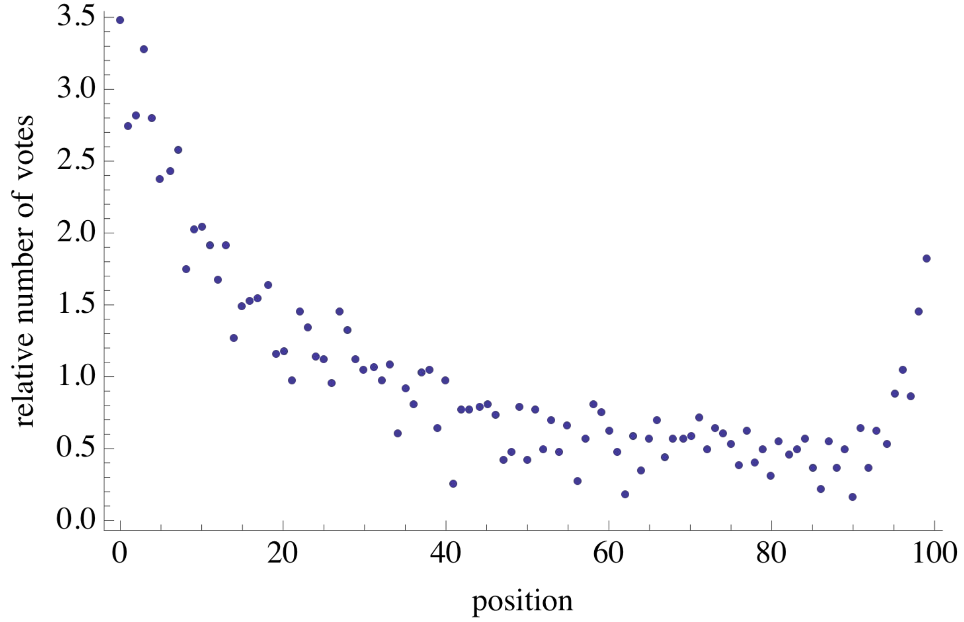
[by recency]

of likes

[Lerman & Hogg (2014) "Leveraging position bias to improve peer recommendation" in *PLoSOne*]

Position bias

All things being equal, items in top list positions receive 4x-5x more attention than those in lower positions

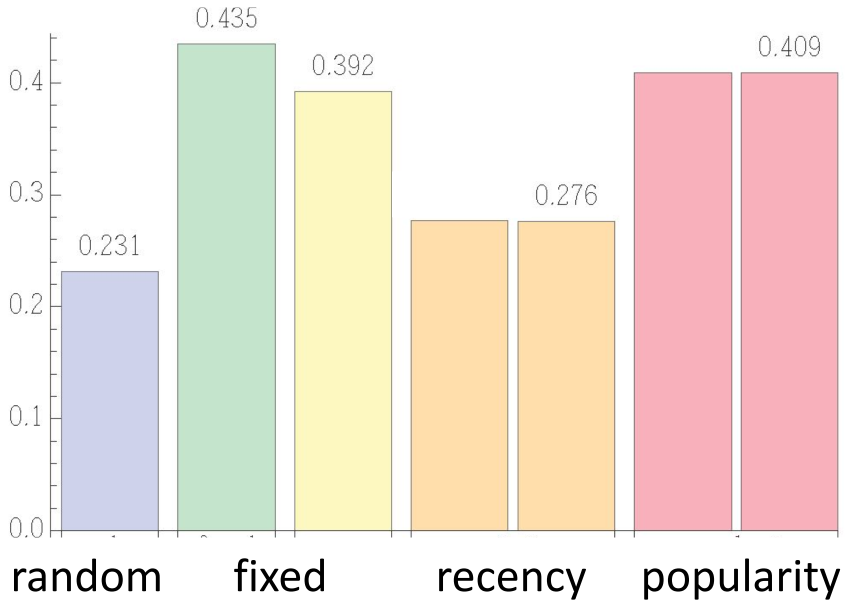


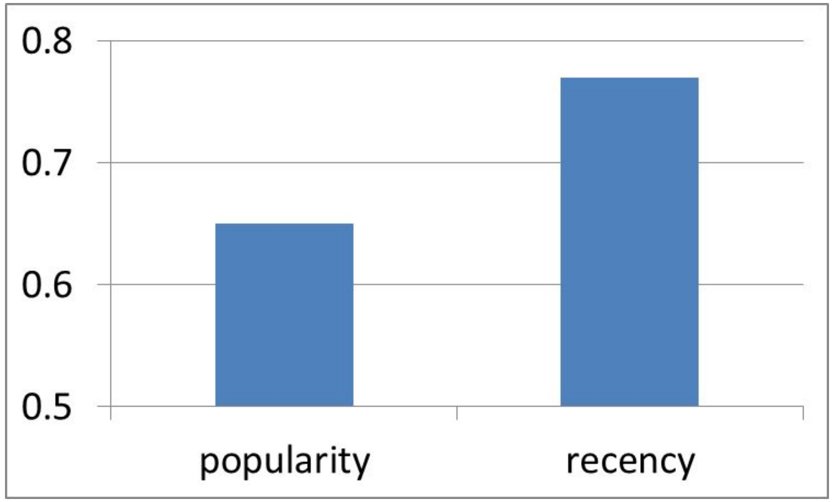
[Lerman & Hogg (2014) "Leveraging position bias to improve peer recommendation" in *PLoSOne*]

Same outcomes as MusicLab

Inequality of popularity: Gini coefficient of the number of likes received

Unpredictability of popularity: Correlation of outcomes across parallel worlds



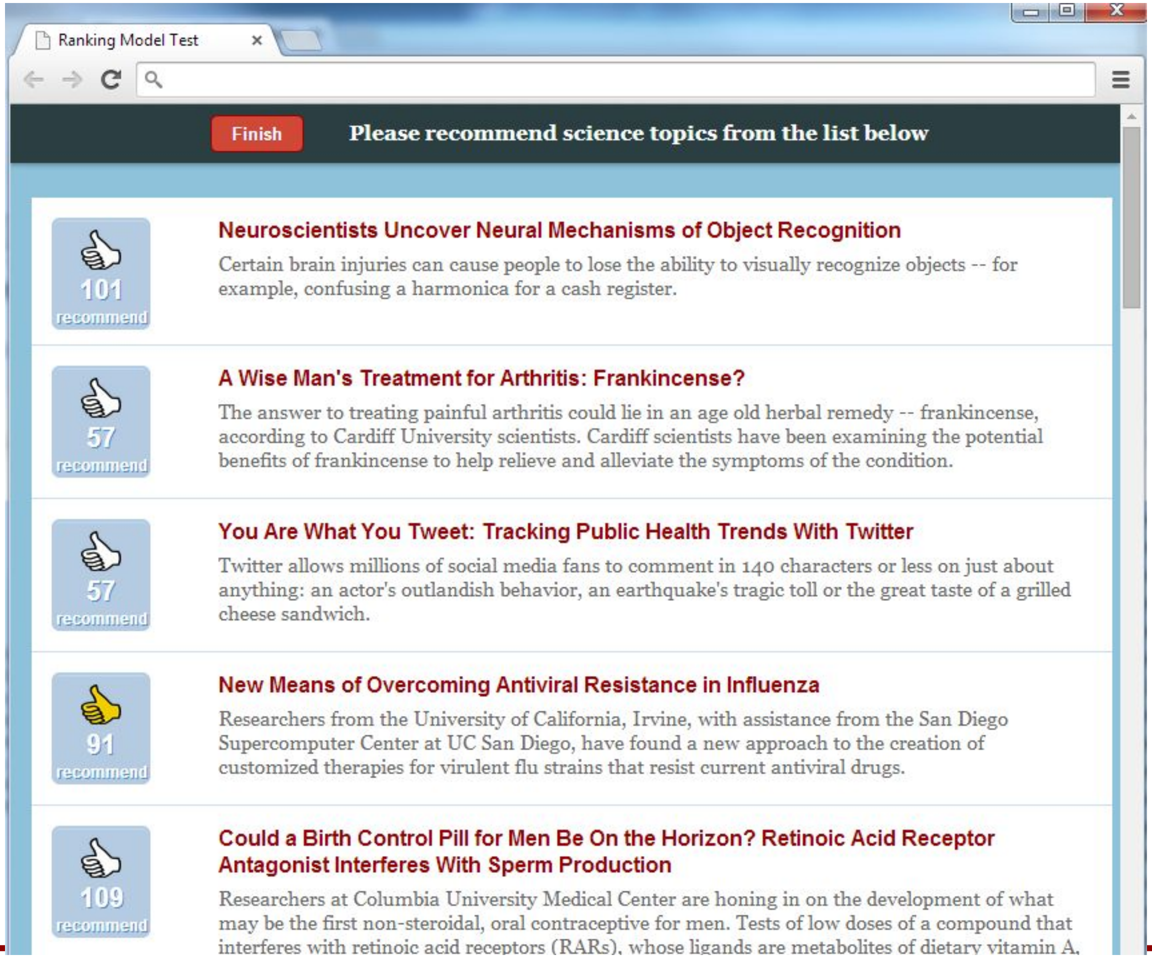


Position bias results in large inequality and unpredictability when ranking items by popularity

[Lerman & Hogg (2014) "Leveraging position bias to improve peer recommendation" in *PLoSOne*]

Measuring social influence bias

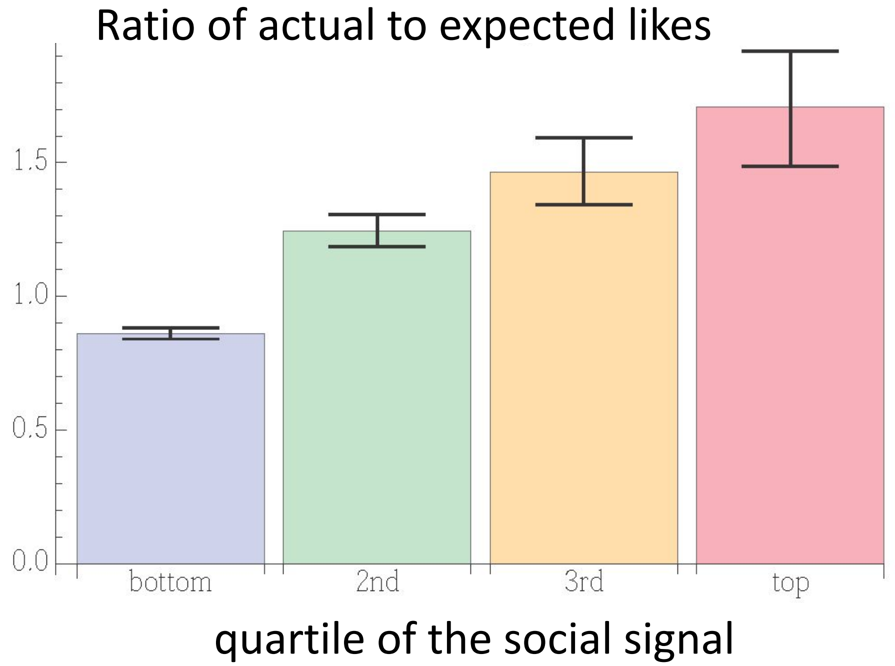
How does "social proof" (the number of likes) affect collective outcomes (popularity)?



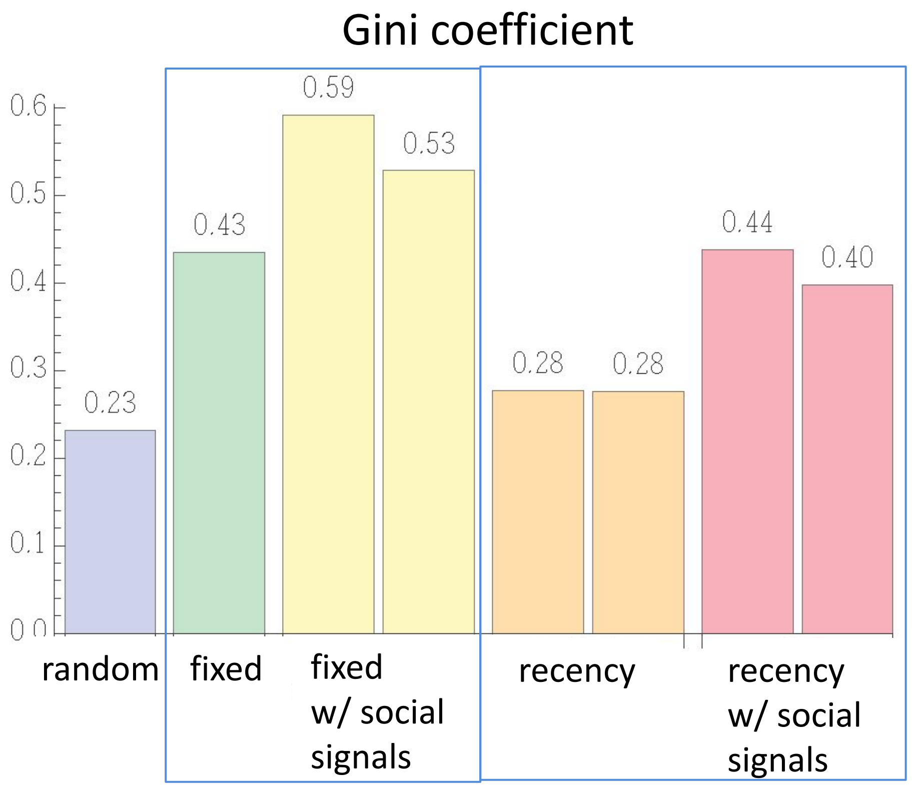
[Hogg & Lerman (2015) "Disentangling the effects of social signals" in *Human Computation Journal*]

Social influence and collective outcomes

Items with higher "social proof" get more attention (after controlling for quality and position)



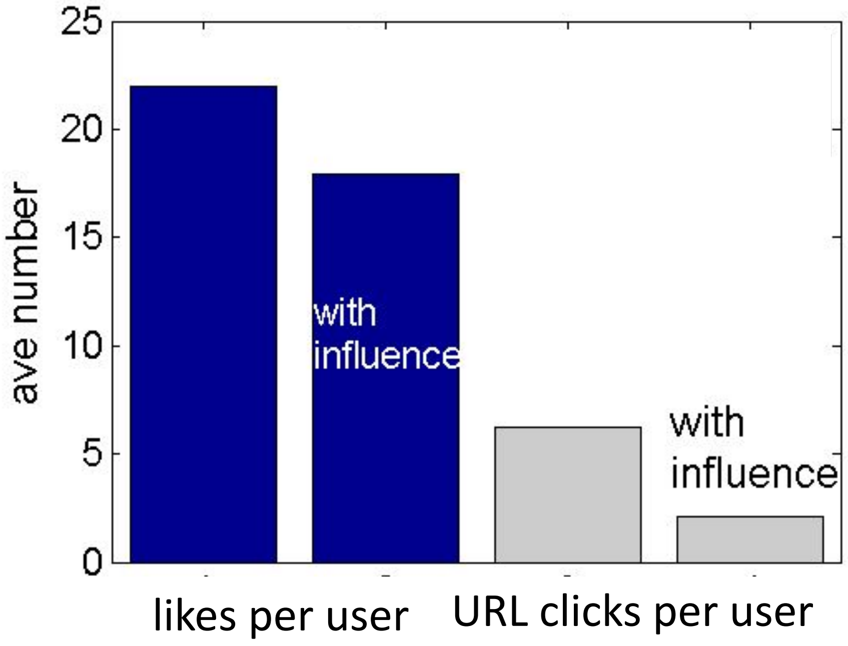
"Social proof" (influence) increases inequality



[Hogg & Lerman (2015) "Disentangling the effects of social signals" in *Human Computation Journal*]

Social influence improves collective efficiency

- **Social influence reduces effort:** people read and like fewer stories with social signals (20% less time)



[Hogg & Lerman (2015) "Disentangling the effects of social signals" in *Human Computation Journal*]

Takeaways

- **Socially-embedded AI algorithms are inherently dynamic**
  - Interactions between people and AI systems produces new data
  - Interactions are predictably skewed
  - … often amplify bias